Joint-Embedding Predictive Architectures and Latent Residual Conditioning for Multivariate Time-Series Anomaly Detection

Andre Williams, Imadeldin Mahgoub, *Senior Member, IEEE,* and Hamid Akbarian

***Abstract*—Deep reconstructive detectors for multivariate time-series anomaly detection minimize observation-space error, fitting stochastic sensor noise that produces heavy-tailed residuals and severe false-alarm flooding under automated threshold calibration. We study how Joint-Embedding Predictive Architectures (JEPAs) condition predictive residuals in a 32-dimensional latent space under distribution shift. We compare an unconstrained single-horizon baseline (TS-JEPA) against Soliton-Reynolds JEPA, a multi-horizon architecture that taps intermediate temporal convolutional stages, subtracts online context batch barycenters, whitens multi-horizon residuals, and scores centered sub-horizon fluctuations via a Lieb–Thirring spectral concentration ratio. Across 33 continuous telemetry streams from seven physical datasets (19 non-Gasoil Heating Loop (GHL) streams at five seeds and 14 GHL streams at three seeds), we calibrate thresholds on chronological held-out nominal data using Streaming Peaks-Over-Threshold (SPOT) at a 0.1% design risk. On the 19 non-GHL streams, latent scoring cuts stream-macro held-out false-positive rates (FPRs) three-fold under retrospective buffered attribution (9.3% to 10.8% versus 30.0% to 30.7% for TimesNet and TranAD) and ten- to eighty-fold under strictly causal streaming (2.4% for TS-JEPA and 0.3% for Soliton-Reynolds JEPA versus 22.2% to 24.3%). Similar reductions hold across all 33 streams (0.3% to 1.5% causal FPR versus 14.2% to 16.1% for reconstructors), and Soliton-Reynolds JEPA achieves the highest buffered precision-recall area under the curve (0.375) and causal matched-FPR event recall (0.701 at 1% FPR). Matched-encoder factorial ablations show that Layer Normalization (LayerNorm) bottlenecks compress residual kurtosis, learned latent projections and multi-horizon barycenter whitening cut calibration-to-holdout drift, short prediction horizons (one to four steps) eliminate emission latency, and hybrid sensor variance monitors recover quiet single-channel faults at a quantified false-alarm cost.**

***Index Terms*—Time-series anomaly detection, Multivariate analysis, Threshold calibration, Extreme value theory, Representation learning.**

I. Introduction

Autonomous anomaly detection in multi-sensor telemetry monitors operational health across aerospace systems, Earth observation, cyber-physical infrastructure, and wearable healthcare [1], [2]. In NASA SMAP and MSL rovers, industrial process loops, and biometric sensor arrays, dozens of channels track power buses, thermal gradients, vibrations, and kinematics [3]. Monitors must identify subsystem faults without overwhelming operators with nuisance alerts.

False-Alarm Flooding Under Automated Threshold Calibration: Operational monitoring requires automated decision rules that convert continuous scores into binary alarms using nominal calibration data alone. Prevailing deep detectors (TimesNet, TranAD, Anomaly Transformer) minimize point-level reconstruction error in raw observation space [4]–[6]. Because raw telemetry contains sensor jitter and scale disparities, observation residuals exhibit heavy-tailed excess kurtosis ($\gamma_2 > 8$) and serial clustering. When paired with Streaming Peaks-Over-Threshold (SPOT) [7] at design risk $q = 10^{-3}$ (0.1%), heavy-tailed residuals suffer calibration breakdown on unseen nominal data, triggering false alarms across 22%–31% of nominal timesteps.

Latent Residual Conditioning and Scope of Claims: Predicting the *summary* of a future window, rather than every raw sample, filters out unforecastable sensor jitter. We study Joint-Embedding Predictive Architectures (JEPAs) [8], which forecast a 32-dimensional latent summary of the next S steps ($S = 64$ by default, or $S \in \{1, 4\}$ for low-latency tripwires) from the preceding 256 steps without synthetic anomaly injection [9]. Alongside an unconstrained single-horizon baseline (`TS-JEPA`), we formulate a multi-horizon fluid-dynamic decomposition model, *Soliton-Reynolds JEPA*. To capture fast onset transients while removing slow baseline drift, *Soliton-Reynolds JEPA* taps intermediate temporal convolutional stages, subtracts online context batch barycenters, and whitens prediction residuals across four nested sub-horizons. It then separates the whitened residuals into mean transport and centered sub-horizon fluctuations, scoring wavepacket coherence via a Lieb–Thirring spectral concentration ratio. Our central contribution is relative false-alarm suppression, reduced calibration-to-holdout drift, and multi-horizon ranking gains rather than solved absolute calibration at $q = 10^{-3}$. We evaluate: (i) how much latent residuals reduce realized false-positive rates (FPRs) and out-of-sample calibration error $|\text{FPR}_{\text{hold}} - q|$ under static, declustered, and rolling calibrators; (ii) how normalization, learned latent projection, multi-horizon whitening, and soliton concentration isolate the mechanism of improvement; (iii) how detection varies across abrupt, gradual, and quiet single-channel faults; and (iv) the trade-off between score-emission latency ($S - 1$) and detection quality.

Manuscript received October 15, 2026. This work was supported by the Department of Electrical Engineering and Computer Science at Florida Atlantic University. No prior conference presentation or thesis material is included in this submission. *(Corresponding author: Andre Williams.)*

Andre Williams, Imadeldin Mahgoub, and Hamid Akbarian are with the Department of Electrical Engineering and Computer Science, Florida Atlantic University, Boca Raton, FL 33431 USA (e-mail: andrewilliam2018@fau.edu; mahgoubi@fau.edu; hakbaria@fau.edu).

Color versions of one or more figures in this article are available online at https://ieeexplore.ieee.org.

A. Contributions

1) **Factorial isolation of tail compression vs. drift reduction:** Across 12 matched-backbone and non-parametric arms (272k–312k online parameters), a layer-normalization (LayerNorm) bottleneck bounds heavy residual tails (γ_2 : $6.10 \rightarrow 0.91$). Moving prediction into a learned latent space halves calibration-to-holdout drift (D_{KS} : $0.376 \rightarrow 0.214$ in `TS-JEPA`), and multi-horizon barycenter whitening in `Soliton-Reynolds JEPA` lowers drift further to $D_{\mathrm{KS}} = 0.163$ (Section VI-D, Table II(c)).
2) **Fewer false alarms, leading multi-horizon ranking, and hybrid quiet-fault recovery:** Across the 19 non-GHL streams, latent scoring cuts held-out FPRs threefold under retrospective buffered attribution (9.3%–10.8% vs. 30.0%–30.7% for TimesNet and TranAD) and ten- to eighty-fold under strictly causal streaming (0.3%–2.4% vs. 22.2%–24.3%). `Soliton-Reynolds JEPA` achieves leading buffered precision-recall area under the curve (PR-AUC 0.375), point-wise F1 (Point-F1 0.188), and causal matched-FPR event recall (0.701 at 1% FPR), and pairing latent prediction with a sensor variance monitor raises quiet single-channel event recall from 0.19–0.28 to 0.42–0.50 at a +15.1% nominal FPR cost (Tables II(c) and III, Figs. 3(b) and 4(b)).
3) **Lieb–Thirring soliton concentration bounds and short-horizon latency control:** Proposition 1 proves that Reynolds-centered multi-horizon fluctuation Gram matrices $\mathbf{G} = \mathbf{E}'\mathbf{E}'^{\top} \in \mathcal{S}_+^M$ have an exact rank deficiency ($\lambda_1 \equiv 0$) and obey the tight scale-invariant bound $\frac{1}{M-1} \leq R_{\mathrm{LT}}(\mathbf{E}') \leq 1$, separating isotropic sub-horizon noise ($R_{\mathrm{LT}} = \frac{1}{M-1}$) from rank-1 soliton wavepackets ($R_{\mathrm{LT}} = 1$). Fixed-backbone ablations show that the multi-horizon whitened envelope drives the ranking and drift gains while the post-hoc soliton multiplier satisfies Two One-Sided Tests (TOST) equivalence without altering Point-F1, and short horizons ($S \in \{1, 4\}$) eliminate the 63-step emission lag (Tables II and III, Fig. 4(a)).
4) **Decomposed calibration error across static, declustered, and adaptive rules:** A refitted permutation decomposition across all 19 non-GHL streams shows that inter-block regime drift accounts for 71.3% of excess causal risk in single-horizon `TS-JEPA` and 78.3%–81.8% in reconstructors (with resolved positive confidence intervals (CIs)). In contrast, inter-block drift vanishes (0.0%, $\mathrm{FPR}_{\mathrm{chr}} = 0.0033$ vs. $\mathrm{FPR}_{\mathrm{blk}} = 0.0037$) in barycenter-centered `Soliton-Reynolds JEPA`, and runs-declustered SPOT and rolling quantiles narrow the gap on step and variance shifts (Table IV).

Roadmap. Section II reviews related work and Section III details the benchmark cohort. Section IV formulates the architectures, Section V defines the protocol, Sections VI and VII report ablations and benchmark results, and Section VIII concludes.

TABLE I
KEY OPERATIONAL AND STATISTICAL TERMS USED THROUGHOUT THE MANUSCRIPT.

Term	Meaning
Latent residual	Gap between the predicted and observed 32-dimensional summary of the next window.
SPOT	Streaming extreme-value rule that sets an alarm threshold from nominal scores alone.
Calibration error	Absolute gap $\lvert\mathrm{FPR}_{\mathrm{hold}} - q\rvert$ between realized held-out nominal FPR and design risk $q{=}10^{-3}$.
Event recall	Fraction of true anomaly segments with at least one alarm.
Buffered	Retrospective scoring that spreads each window score over its S steps.
Causal	Scoring that uses only past data and assigns each score to its final step $\tau = t + S - 1$.
Calibration drift	Shift between calibration and later nominal residuals, measured by Kolmogorov–Smirnov distance.
Excess kurtosis	Tail heaviness of the residual distribution; large values mean rare, extreme spikes.
Soliton ratio	Lieb–Thirring concentration $R_{\mathrm{LT}} \in [\frac{1}{M-1}, 1]$ of centered sub-horizon fluctuations.

II. RELATED WORK

Observation-Space Forecasting and Reconstructive Detectors: Deep time-series anomaly detectors predominantly forecast sensor readings or reconstruct input windows in raw observation space, flagging timesteps where point-level error $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$ exceeds a threshold. Autoregressive long short-term memory (LSTM) networks with non-parametric dynamic thresholding [3] model sequential dependencies directly on spacecraft channels. Recent Transformer and spectral reconstructors introduce specialized attention and frequency modules: TranAD [6] combines adversarial two-phase training with self-conditioning sub-series attention; Anomaly Transformer [4] contrasts prior Gaussian kernel associations against series-level self-attention; and TimesNet [5] folds 1D sequences into 2D tensors along dominant Fast Fourier Transform (FFT) periods for 2D convolutional modeling. Because point-level losses penalize error across every channel equally, reconstructors fit stochastic sensor jitter and environmental fluctuations. In high-dimensional streams ($K \geq 25$), summing raw channel errors accumulates heavy-tailed (*leptokurtic*, $\gamma_2 > 0$) residual spikes that destabilize extreme-value threshold fitting under baseline shift [10].

Contrastive and Joint-Embedding Predictive Architectures: Self-supervised representation methods map windows into a compact latent space. Contrastive detectors construct positive and negative pairs: TS2Vec [11] maximizes contextual agreement across hierarchical sub-series; Neural Contextual Anomaly Detection (NCAD) [9] contrasts full-window and context-only embeddings using injected synthetic anomalies; and DCdetector [12] employs dual patch-wise and in-patch attention without reconstruction. However, contrastive detectors depend on whether synthetic perturbations or patch discrepancies match test-time fault morphology (0.058 and 0.070 macro Point-F1 for NCAD and DCdetector vs. 0.160 for latent prediction across 11 datasets, Table III(d)). Joint-Embedding Predictive Architectures (JEPAs) [8], [13], [14] eliminate observation decoding and negative pairs by encoding context z_{ctx} and predicting the target embedding z_{tgt} produced by a separate target encoder. Collapse is prevented via exponential moving average (EMA) target updates [15] or variance-covariance penalties such as VICReg [16], Barlow Twins [17], and batch whitening [18]. How latent predictive residuals behave under unsupervised extreme-value threshold calibration remains unexplored.

Fluid-Dynamic Reynolds Decomposition and Spectral Concentration Priors: Multi-sensor telemetry reflects physi-

cal dynamical trajectories [19]. Neural ordinary differential equations (ODEs) [20], Physics-Informed Neural Networks (PINNs) [21], and Hamiltonian or Lagrangian networks [22], [23] enforce differential conservation laws on physical coordinates, but require known governing equations and suffer from stiff gradient pathologies on uncalibrated sensor arrays [24]. Koopman and operator-theoretic models [25]–[29] lift nonlinear dynamics into learned observable spaces. Rather than imposing partial differential equations on raw sensors, we draw on Reynolds decomposition from turbulence modeling to separate multi-horizon latent predictions into a mean trajectory and centered sub-horizon fluctuations. We then measure wavepacket localization across prediction horizons via a Lieb–Thirring spectral concentration ratio on the centered fluctuation Gram matrix.

Extreme-Value Threshold Calibration and Evaluation Rigor: Streaming Peaks-Over-Threshold (SPOT) [7] fits a Generalized Pareto Distribution (GPD) to score exceedances above quantile u to extrapolate a threshold at design risk q. However, continuous telemetry violates the classical independent and identically distributed (i.i.d.) assumption through short-range serial correlation (quantified by the Ferro–Segers extremal index [30]) and inter-block regime drift. Because "point-adjusted" F1 severely inflates scores [10], [31], [32], we evaluate all models using unadjusted Point-F1, PR-AUC, causal vs. buffered attribution, and matched realized false-alarm operating curves.

III. Multi-Sensor Internet-of-Things and Remote Sensing Benchmark Suite

Our benchmark suite comprises 11 datasets spanning 45 continuous telemetry streams from NASA spacecraft archives [3], the University of California, Irvine (UCI) Machine Learning Repository [33], [34], TimeEval [10], and mTSBench [35]. Thirty-three streams enter the primary multi-seed held-out protocol; 12 are reserved with explicit accounting below. The suite spans four physical domains and separates natural operational or physiological anomalies from testbed or simulated faults: *spacecraft and solar telemetry* (six SMAP streams, K=25, 7 anomaly events; four primary and two short-nominal MSL rover streams, K=55, 6 primary events; and SWAN-SF solar magnetograms [36], K=38, 6,615 flare events, all natural operational faults); and *wearable kinematics* (six Daphnet Freezing-of-Gait (FoG) streams [33] at 64 Hz, K=9, 72 natural gait-freezing episodes, where S=64 equals a 1 s horizon). In *industrial processes* and *smart infrastructure*, the suite covers 14 Gasoil Heating Loop (GHL) streams (K=16, 26 testbed valve and heater faults), the Genesis pick-and-place testbed (K=18, 2 actuator faults), GECCO water chemistry at 1-minute resolution (K=9, 37 natural hydrological anomalies), Metro I-94 hourly highway traffic (K=5, 25 natural traffic events), Room-Occupancy sensing [37] (K=5), CalIt2 building flow [34] (K=2), and six CICIDS network streams (K=72, simulated attack flows). Cohort 3 (Section VI-B) separately injects six controlled single-channel faults into six real streams.

Each stream provides nominal training sequences ($N_{\text{tr}} \in [2{,}074, 120{,}000]$ across the 33 primary streams) and labeled anomaly test sequences ($N_{\text{test}} \in [2{,}158, 114{,}001]$). Channels are standardized by training statistics with $\boldsymbol{\sigma}$ floored at 10^{-2} against flatlines, matching standardized archive subsets in prior benchmarks [3], [10].

Each nominal sequence is split chronologically into training (65%), nominal calibration (17.5%), and held-out nominal evaluation (17.5%) blocks separated by an unobserved 64-step isolation gap (Section V-A). Inclusion in the primary multi-seed protocol required two prespecified criteria: (i) at least $N_{\text{cal}} \geq 500$ calibration windows ($M_{\text{val}} = \lceil 0.35\, T_{\text{nom}} \rceil - 319 \geq 1{,}000$, requiring $T_{\text{nom}} \geq 3{,}769$), and (ii) tractable stride-1 multi-seed runtime ($K \leq 55$, $N_{\text{test}} \leq 1.2 \times 10^5$). Exactly 33 streams across seven datasets satisfy both criteria (Table III(c)). Among the 12 reserved streams, five have nominal sequences too short to yield 500 calibration windows: CalIt2 `traffic` (T_{nom}=940, N_{cal}=4), MSL `C-2` (T_{nom}=764, N_{cal}=0), MSL `D-16` (T_{nom}=1,451, N_{cal}=94), and Room-Occupancy `default` and `1` (T_{nom}=1,073 and 1,492; N_{cal}=28 and 101). Six CICIDS network streams (`0,2,3,5,6,7`, K=72, $N_{\text{test}} \in [1.36 \times 10^5, 4.15 \times 10^5]$) exceed the stride-1 multi-seed runtime envelope, and GECCO `water_quality` (K=9, 1 held-out event) is used in Cohorts 3–4. All 11 datasets and 130 archive streams appear in the secondary single-seed cross-check (Table III(d)).

Two caveats bound external validity: (i) SMAP and MSL contain documented labeling flaws [32]; and (ii) streams within SMAP, MSL, Daphnet, and GHL share hardware platforms (N=7 dataset clusters). We therefore report Dataset-Macro means, between-stream and within-stream seed standard deviations (SDs), zero-tuning leave-one-dataset-out (LODO) sensitivity, and dataset-clustered permutation tests (Table III(c)).

IV. Methodology: Joint-Embedding Predictive Architectures and Soliton-Reynolds Decomposition

A. Problem Formulation and Latent Space Prediction

Consider a multivariate telemetry stream $\mathbf{X} \in \mathbb{R}^{T \times K}$ recording K sensor channels over T timesteps. As shown in Fig. 1 (Stage 1), a sliding window of length $W = C + S = 320$ advances at stride 1 and splits the series at index t into: (i) a *context window* $\mathbf{x}_{\text{ctx}} = \mathbf{X}[t - C : t, :] \in \mathbb{R}^{C \times K}$ ($C = 256$), establishing the nominal regime; and (ii) a *suspect target horizon* $\mathbf{x}_{\text{tgt}} = \mathbf{X}[t : t + S, :] \in \mathbb{R}^{S \times K}$ ($S = 64$, up to decision timestamp $\tau = t + S - 1$), inspected for trajectory departures.

Unlike reconstructive detectors that minimize pointwise observation error $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$ and inherit leptokurtic sensor noise (Sections I and II), we shift prediction into a 32-dimensional latent space ($D = 32$, Fig. 1) using a Joint-Embedding Predictive Architecture (JEPA) [8]. An online context encoder E_θ maps $\mathbf{x}_{\text{ctx}}$ into $z_{\text{ctx}} \in \mathbb{R}^D$, while a target encoder E_ϕ maps $\mathbf{x}_{\text{tgt}}$ into $z_{\text{tgt}} \in \mathbb{R}^D$. To prevent constant-vector collapse, target weights ϕ are updated as an EMA of online weights θ, $\phi_k \leftarrow m_k \phi_{k-1} + (1 - m_k)\theta_k$ with cosine momentum $m_k \in [0.995, 0.9995]$ [13], [15].

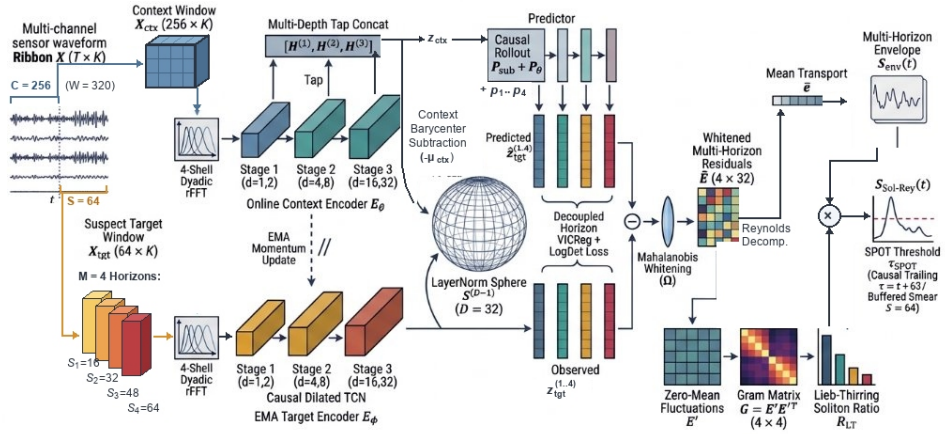


Fig. 1. Four-stage `Soliton-Reynolds JEPA` architecture. (1) Multiscale windowing splits telemetry at index t into context ($\mathbf{x}_{\text{ctx}} \in \mathbb{R}^{256\times K}$) and target ($\mathbf{x}_{\text{tgt}} \in \mathbb{R}^{64\times K}$) blocks processed by a 4-shell dyadic real fast Fourier transform (rFFT) front-end and causal dilated temporal convolutional network (TCN). (2) Dual online (E_θ) and exponential moving average (EMA) target (E_ϕ) encoders tap all $L = 3$ TCN stages and apply synchronized barycenter centering onto $\sqrt{D}\mathbb{S}^{D-1}$ ($D = 32$). (3) Nested causal multi-horizon rollout forecasts $M = 4$ sub-horizons ($S_m \in \{16, 32, 48, 64\}$) under decoupled horizon Variance-Invariance-Covariance Regularization (VICReg) and log-determinant whitening. (4) Mahalanobis residual whitening, Lieb–Thirring soliton concentration modulation, and coordinate-saliency-gated Streaming Peaks-Over-Threshold (SPOT) calibration under causal trailing ($\tau = t + 63$) and buffered smear mappings.

B. Latent Predictor Architectures: TS-JEPA and Soliton-Reynolds JEPA

We evaluate two nested latent predictors sharing the same multiscale encoder backbone and coordinate saliency gate: an unconstrained single-horizon baseline (`TS-JEPA`) and a multi-horizon fluid-dynamic decomposition model (`Soliton-Reynolds JEPA`).

1) Unconstrained TS-JEPA Baseline: In `TS-JEPA`, a two-layer multi-layer perceptron (MLP) predictor $P_\theta : \mathbb{R}^D \to \mathbb{R}^D$ maps z_{ctx} to a single full-horizon prediction $\hat{z}_{\text{tgt}} = P_\theta(z_{\text{ctx}})$ trained with standard single-horizon predictive alignment and VICReg variance-covariance penalties [16] ($\gamma_{\text{var}} = 1.0$). Terminal LayerNorm bounds latent representations on the sphere of radius $\sqrt{D}$ ($\|z\|_2 = \sqrt{D}$) and compresses nominal residual kurtosis relative to observation-space reconstructors. However, single-horizon encoding pools the entire $S = 64$ target window into one terminal TCN state, smoothing out short-lived sub-horizon transients and leaving cross-horizon residual correlations unwhitened.

2) Soliton-Reynolds JEPA (Multi-Horizon Reynolds Decomposition and Lieb–Thirring Concentration): **Physical Motivation and Structural Design.** In fluid turbulence, Reynolds decomposition separates a velocity field into its time-averaged mean flow and zero-mean fluctuations, while dispersive wave theory characterizes localized, shape-preserving wavepackets (*solitons*) by spectral concentration of the fluctuation operator. `Soliton-Reynolds JEPA` brings both principles into latent time-series prediction through two architectural stages before residual scoring:

1) *Multi-depth TCN functional hierarchy taps and synchronized barycenter centering:* Standard TCN encoders read only the deepest dilated block, where wide receptive fields attenuate sharp onset transients. In `Soliton-Reynolds JEPA`, a multi-depth tap head extracts intermediate feature maps $\mathbf{H}^{(\ell)} \in \mathbb{R}^{B\times T_w \times d_{\text{model}}}$ ($d_{\text{model}} = 64$) across all $L = 3$ causal dilated TCN stages, concatenating the temporal mean and terminal state of each stage with the base encoder output into $\mathbf{h}_{\text{cat}} \in \mathbb{R}^{B\times(2Ld_{\text{model}}+D)}$. A two-layer residual projection maps $\mathbf{h}_{\text{cat}}$ onto the unit sphere $\mathbf{u} \in \mathbb{S}^{D-1}$. To eliminate common-mode baseline drift between context and target windows, the online context batch barycenter $\bar{\boldsymbol{\mu}}_{\text{ctx}} = \text{sg}\big(B^{-1}\sum_{b=1}^{B}\mathbf{u}_{\text{ctx}}^{(b)}\big)$ is subtracted from both online context and EMA target spherical embeddings prior to final LayerNorm projection.
2) *Nested causal multi-horizon rollout and decoupled horizon VICReg:* Rather than predicting only the full $S = 64$ window, we partition $\mathbf{x}_{\text{tgt}}$ into $M = 4$ nested causal prefix windows $\mathbf{x}_{\text{tgt}}^{(m)} = \mathbf{x}_{\text{tgt}}[0 : S_m, :]$ with lengths $S_m = (m/M)S \in \{16, 32, 48, 64\}$. The EMA target encoder encodes each prefix into $z_{\text{tgt}}^{(m)} \in \mathbb{R}^D$. From z_{ctx}, a sub-horizon predictor P_{sub} rolls out intermediate predictions $\hat{z}_{\text{tgt}}^{(m)} = \text{LN}\big(P_{\text{sub}}(\hat{z}_{\text{tgt}}^{(m-1)} + \mathbf{p}_m)\big)$ for $m \in \{1, \ldots, M-1\}$ ($\hat{z}_{\text{tgt}}^{(0)} \equiv z_{\text{ctx}}$) using learnable horizon embeddings $\mathbf{p}_m \in \mathbb{R}^D$, while the primary predictor P_θ forecasts the terminal horizon $\hat{z}_{\text{tgt}}^{(M)} = \text{LN}\big(P_\theta(z_{\text{ctx}} + \mathbf{p}_M)\big)$. With horizon weights $w_m = \frac{1}{2(M-1)}$ for $m < M$ and $w_M = \frac{1}{2}$, the training objective is:

$$\begin{aligned}\mathcal{L}_{\text{Sol-Rey}} = &\sum_{m=1}^{M} w_m \Big(2 - \frac{2}{B}\sum_{b=1}^{B}\big\langle \bar{z}_{\text{pred}}^{(b,m)}, \text{sg}\big(\bar{z}_{\text{tgt}}^{(b,m)}\big)\big\rangle\Big) \\ &+ \frac{1}{M}\sum_{m=1}^{M}\big(\mathcal{L}_{\text{var}}(\hat{z}_{\text{tgt}}^{(m)}) + 0.1\,\mathcal{L}_{\text{cov}}(\hat{z}_{\text{tgt}}^{(m)})\big) \\ &+ 0.05\,\mathcal{L}_{\text{logdet}}(z_{\text{ctx}}),\end{aligned} \tag{1}$$

where $\bar{z} = z/\|z\|_2$, $\mathcal{L}_{\text{var}}(z) = D^{-1}\sum_{d=1}^{D}\max(0, 1 - \sigma_d(z))$ enforces unit per-coordinate standard deviation, $\mathcal{L}_{\text{cov}}(z) = D^{-1}\|\,\text{Cov}(z) - \text{diag}(\text{Cov}(z))\|_F^2$ decorrelates coordinates within each horizon m independently, and $\mathcal{L}_{\text{logdet}}(z_{\text{ctx}}) = -D^{-1}\log\det\big(\text{Cov}(\bar{z}_{\text{ctx}})\sqrt{D} + 10^{-2}\mathbf{I}_D\big)$ prevents dimensional collapse (achieving $\text{RankMe}(z_{\text{tgt}}) = 19.8$ vs. 11.6 for single-horizon baselines).

Reynolds Fluctuation Gram Matrix and Lieb–Thirring Soliton Concentration. Physically, nominal background fluctuations disperse energy across multiple uncorrelated sub-horizons, whereas an emerging fault wavepacket concentrates

fluctuation energy into a single dominant rank-1 directional mode across horizons. To formalize this distinction, let $\mathbf{E} = [\mathbf{e}_1, \ldots, \mathbf{e}_M]^\top \in \mathbb{R}^{M\times D}$ stack the M prediction residuals $\mathbf{e}_m = z_{\text{tgt}}^{(m)} - \hat{z}_{\text{tgt}}^{(m)}$. We fit the empirical shrinkage precision matrix $\mathbf{\Omega} = (\hat{\mathbf{\Sigma}}_{\text{res}} + 10^{-3}\mathbf{I}_{MD})^{-1/2} \in \mathcal{S}_{++}^{MD}$ and mean $\bar{\boldsymbol{\mu}}_{\text{res}} \in \mathbb{R}^{MD}$ on nominal training windows to obtain Mahalanobis-whitened residuals $\tilde{\mathbf{e}} = \mathbf{\Omega}(\text{vec}(\mathbf{E}) - \bar{\boldsymbol{\mu}}_{\text{res}}) \in \mathbb{R}^{MD}$, reshaped into $\tilde{\mathbf{E}} = [\tilde{\mathbf{e}}_1, \ldots, \tilde{\mathbf{e}}_M]^\top \in \mathbb{R}^{M\times D}$. Applying Reynolds decomposition across the M horizons separates $\tilde{\mathbf{E}}$ into its horizon-mean transport $\bar{\mathbf{e}} = M^{-1}\sum_{m=1}^{M}\tilde{\mathbf{e}}_m \in \mathbb{R}^D$ and zero-mean fluctuations $\mathbf{E}' = \tilde{\mathbf{E}} - \mathbf{1}_M\bar{\mathbf{e}}^\top \in \mathbb{R}^{M\times D}$. From the $M \times M$ fluctuation Gram matrix $\mathbf{G} = \mathbf{E}'\mathbf{E}'^\top \in \mathcal{S}_+^M$ with ordered eigenvalues $0 \le \lambda_1 \le \lambda_2 \le \cdots \le \lambda_M$, we define the Lieb–Thirring spectral concentration ratio of order $\gamma = 1$:

$$R_{\text{LT}}(\mathbf{E}') = \frac{\lambda_M^{\gamma+1/2}}{\sum_{m=1}^{M}\lambda_m^{\gamma+1/2} + \epsilon_{\text{LT}}}, \quad \epsilon_{\text{LT}} = 10^{-12}. \tag{2}$$

Proposition 1 establishes the exact algebraic bounds and radial scale invariance of $R_{\text{LT}}(\mathbf{E}')$.

Proposition 1 (Lieb–Thirring Soliton Concentration Bounds, Centered Gram Rank Deficiency, and Scale Invariance)**.** *Let $\mathbf{E}' = (\mathbf{I}_M - M^{-1}\mathbf{1}_M\mathbf{1}_M^\top)\tilde{\mathbf{E}} \in \mathbb{R}^{M\times D}$ denote the Reynolds-centered multi-horizon fluctuation matrix for $M \ge 2$ and $\gamma > -1/2$. In the exact limit $\epsilon_{LT} \to 0^+$ for any non-zero fluctuation $\mathbf{E}' \neq \mathbf{0}$: (i)* Centered Gram nullspace and effective rank $M-1$: *Because $\mathbf{E}'^\top\mathbf{1}_M = \mathbf{0}$, the constant vector $\mathbf{1}_M$ lies in* $\ker(\mathbf{G})$*, forcing $\lambda_1(\mathbf{G}) \equiv 0$ identically so that* $\text{rank}(\mathbf{G}) \le \min(M-1, D)$*. (ii)* Sharp isotropy lower bound and soliton upper bound: *The concentration ratio satisfies the tight two-sided inequality*

$$\frac{1}{M-1} \le R_{LT}(\mathbf{E}') \le 1, \tag{3}$$

where the lower bound $\frac{1}{M-1}$ ($= \frac{1}{3}$ for $M = 4$) is attained if and only if fluctuation energy is distributed isotropically across $M-1$ orthogonal sub-horizon modes ($\lambda_2 = \cdots = \lambda_M > 0$), and the upper bound 1 is attained if and only if $\text{rank}(\mathbf{E}') = 1$ *($\lambda_2 = \cdots = \lambda_{M-1} = 0 < \lambda_M$), corresponding to a coherent rank-1 soliton wavepacket. (iii)* Exact radial scale invariance: *For any scalar $c > 0$, $R_{LT}(c\mathbf{E}') = R_{LT}(\mathbf{E}')$, rendering R_{LT} invariant to homogeneous amplitude rescaling.*

Proof. By construction, $\mathbf{G}\mathbf{1}_M = \mathbf{E}'(\mathbf{E}'^\top\mathbf{1}_M) = \mathbf{0}$, so $\lambda_1 = 0$ is always an eigenvalue of $\mathbf{G} \succeq \mathbf{0}$, leaving at most $M-1$ positive eigenvalues $\lambda_2 \le \cdots \le \lambda_M$ with $\lambda_M > 0$ since $\mathbf{E}' \neq \mathbf{0}$. Because $x \mapsto x^{\gamma+1/2}$ is strictly increasing on $x \ge 0$ for $\gamma > -1/2$, $\lambda_m^{\gamma+1/2} \le \lambda_M^{\gamma+1/2}$ for all $m \in \{2, \ldots, M\}$. Summing over the $M-1$ remaining modes gives $\lambda_M^{\gamma+1/2} \le \sum_{m=2}^{M}\lambda_m^{\gamma+1/2} \le (M-1)\lambda_M^{\gamma+1/2}$. Dividing through by $\sum_{m=2}^{M}\lambda_m^{\gamma+1/2}$ yields (3), with equality on the left iff $\lambda_2 = \cdots = \lambda_M$ and on the right iff $\lambda_2 = \cdots = \lambda_{M-1} = 0$. Finally, scaling $\mathbf{E}' \mapsto c\mathbf{E}'$ maps $\mathbf{G} \mapsto c^2\mathbf{G}$ and $\lambda_m^{\gamma+1/2} \mapsto c^{2\gamma+1}\lambda_m^{\gamma+1/2}$, canceling in numerator and denominator of (2) as $\epsilon_{\text{LT}} \to 0^+$. □

C. Anomaly Scoring and Operational Detection Latency

At test time, windows are scored at stride 1. Let $g_d : \mathbb{R}^d \to \mathbb{R}_{\ge 0}$ denote the dimension-agnostic coordinate saliency gate defined in (6). Unconstrained `TS-JEPA` scores the single-horizon residual $\mathbf{e}(t) = z_{\text{tgt}}(t) - \hat{z}_{\text{tgt}}(t)$ via $\mathcal{S}_{\text{TS}}(t) = g_D(\mathbf{e}(t))$. In `Soliton-Reynolds JEPA`, we combine the joint MD-dimensional whitened residual gate $g_{MD}(\tilde{\mathbf{e}}(t))$ with the peak sub-horizon gate $\max_{1\le m\le M} g_D(\tilde{\mathbf{e}}_m(t))$ into the multi-horizon envelope $\mathcal{S}_{\text{env}}(t)$, and modulate it by a one-sided median absolute deviation (MAD)-standardized soliton concentration gain and nominal manifold distance gain:

$$\mathcal{S}_{\text{env}}(t) = \sqrt{\frac{1}{2}\Big(g_{MD}\big(\tilde{\mathbf{e}}(t)\big)^2 + \max_{1\le m\le M} g_D\big(\tilde{\mathbf{e}}_m(t)\big)^2\Big)}, \tag{4}$$

$$\mathcal{S}_{\text{Sol-Rey}}(t) = \mathcal{S}_{\text{env}}(t)\Big(1 + \alpha_{\text{sol}}\max\big(0, \tanh(0.5\, z_{\text{LT}}(t))\big)\Big) \times \Gamma_{\text{bank}}(t), \tag{5}$$

where $\alpha_{\text{sol}} = 0.12$ and $z_{\text{LT}}(t) = (R_{\text{LT}}(\mathbf{E}'(t)) - m_{\text{LT}})/(1.4826\,\text{MAD}_{\text{LT}})$ standardizes R_{LT} against its nominal training median m_{LT} and MAD MAD_{LT}. The factor $\Gamma_{\text{bank}}(t) = 1 + 0.15\tanh(0.5\,\tilde{d}_{\text{core}}(t)) + 0.10\tanh(0.5\,\tilde{d}_{\text{null}}(t))$ applies a bounded gain from MAD-normalized k-nearest-neighbor coreset distance $\tilde{d}_{\text{core}}$ ($N_{\text{core}} = 64, k = 5$) and minor principal-component null-space energy $\tilde{d}_{\text{null}}$ ($d_{\text{null}} = 8$) fitted on nominal training windows. Because $\max(0, \tanh(0.5\, z_{\text{LT}}))$ is one-sided, nominal windows with below-median spectral concentration ($z_{\text{LT}} \le 0$) receive zero amplification. Setting $\alpha_{\text{sol}} = 0$ and $\Gamma_{\text{bank}} \equiv 1$ recovers the unmodulated multi-horizon envelope control $\mathcal{S}_{\text{env}}(t)$ (`Soliton-Reynolds` (α_{sol}=0)).

Three-Part Operational Causality Framework: We formalize online evaluation under three criteria (Fig. 2):

(a) **Data causality ($\mathcal{F}_\tau$-measurability):** At decision timestamp $\tau = t + S - 1$, the detector observes only $\mathbf{X}[t-C : \tau, :]$; neither encoder accesses observations beyond τ.

(b) **Score-emission latency ($S - 1 = 63$ steps) vs. fault-detection delay (Δt):** Because z_{tgt} summarizes suspect observations $\mathbf{X}[t : t+S, :]$ spanning sample indices $[t, t+S-1]$, score s_t is emitted at the trailing edge $\tau = t + S - 1$ (Fig. 2, Row 3), creating a structural emission lag of $S - 1 = 63$ steps (0.98 s on 64 Hz Daphnet, 10.5 min on 0.1 Hz SMAP/MSL/GHL, 63 min on GECCO, and 63 h on hourly Metro). This lag is distinct from empirical fault-detection delay $\Delta t = t_{\text{alarm}} - t^*$ from onset t^* to the first alarm $t_{\text{alarm}} \ge t^*$. Shortening the horizon to $S \in \{1, 4\}$ eliminates emission latency for rapid tripwires (Section VII-D).

(c) **Strictly causal trailing vs. retrospective buffered attribution:** *Causal trailing attribution* assigns s_t only to $\tau = t+S-1$ for zero-lookahead streaming. *Retrospective buffered smear attribution* averages s_t across all suspect timestamps $[t, t+S-1]$, introducing fixed-lag smoothing. Both regimes are reported separately throughout (Table III(a),(b)).

D. Shared Architectural Components and Thresholding

1) Coordinate Saliency Gating and Scale Equivariance: To amplify sparse departures in high-dimensional representations while mitigating channel drowning, we define the coordinate

saliency gate $g_d : \mathbb{R}^d \to \mathbb{R}_{\geq 0}$ for any vector $\mathbf{v} \in \mathbb{R}^d$:

$$g_d(\mathbf{v}) = (1-\alpha)\|\mathbf{v}\|_2 + \alpha\sqrt{d\sum_{i=1}^{d} w_i v_i^2}, \qquad w_i = \frac{\exp\big(|v_i|/(\tau_g \sigma_{\mathbf{v}})\big)}{\sum_{j=1}^{d}\exp\big(|v_j|/(\tau_g\sigma_{\mathbf{v}})\big)}, \quad \sigma_{\mathbf{v}} = \frac{\|\mathbf{v}\|_2}{\sqrt{d}} + \epsilon, \tag{6}$$

with $\alpha = 0.5$, $\tau_g = 0.5$, and $\epsilon = 10^{-6}$. When $\mathbf{v} = \mathbf{0}$, $g_d(\mathbf{0}) = 0$. For non-zero $\mathbf{v}$, rescaling $\mathbf{v} \mapsto c\mathbf{v}$ with $c > 0$ yields $\sigma_{c\mathbf{v}} = c\frac{\|\mathbf{v}\|_2}{\sqrt{d}} + \epsilon = c\sigma_{\mathbf{v}} + (1-c)\epsilon$. In the operational regime where $\|\mathbf{v}\|_2 \gg \epsilon$, the softmax weights w_i are scale-invariant up to an $\mathcal{O}(\epsilon/\|\mathbf{v}\|_2)$ perturbation, satisfying approximate degree-1 homogeneity $g_d(c\mathbf{v}) \approx c\, g_d(\mathbf{v})$ and recovering exact degree-1 homogeneity in the limit $\epsilon \to 0^+$. Input channels use uniform weights in the deployed pipeline (Section VI-B).

2) Multiscale Frequency Front-End: For multivariate streams ($K > 1$), a causal depthwise 1D convolutional stem filters each channel prior to 1×1 mixing ($K \leq 1$ uses an identity bypass). Each projected channel is split across $J = 4$ smooth dyadic shells on a 256-point rFFT grid normalized into an exact amplitude partition of unity ($\sum_{j=1}^{J} \Psi_j[\omega_m] \equiv 1$ within 1.2×10^{-7}), processed by causal 1D convolutions (kernels $\{15, 9, 5, 3\}$), and recombined via a softmax scale router. Because smooth Fourier shells overlap at band edges ($\sum \Psi_j \equiv 1$ rather than $\sum \Psi_j^2 \equiv 1$), this acts as a multiscale front-end rather than an orthogonal Parseval wavelet bank (Table II(b)).

3) Extreme-Value Threshold Calibration (SPOT): To convert scores $\mathcal{S}(t)$ into binary alarms $\hat{y}_t = \mathbb{I}(\mathcal{S}(t) > \tau_{\text{SPOT}})$ from nominal calibration data alone, SPOT [7] fits exceedances $Y = \mathcal{S}(t) - u > 0$ above the 98th calibration percentile $u = p_{98}$ to a Generalized Pareto Distribution (GPD) with shape ξ and scale σ via Grimshaw maximum-likelihood estimation [38] with bounded L-BFGS-B fallback ($\xi \in [-0.9, 10]$). Inverting the GPD tail at target risk $q = 10^{-3}$ yields:

$$\tau_{\text{SPOT}} = \begin{cases} u + \frac{\sigma}{\xi}\left[\left(\frac{qN_{\text{cal}}}{N_u}\right)^{-\xi} - 1\right], & |\xi| \geq 10^{-5}, \\ u - \sigma \ln\left(\frac{qN_{\text{cal}}}{N_u}\right), & |\xi| < 10^{-5}, \end{cases} \tag{7}$$

where the lower branch is the continuous exponential-tail limit as $\xi \to 0$, evaluated directly to avoid division by zero. Here N_{cal} and N_u count post-attribution score timestamps on the calibration block. Primary tables use standard SPOT; Section VII-F compares parametric SPOT against non-parametric empirical quantiles, runs-declustered GPD fitting, and rolling thresholds.

V. Experimental Protocol and Evaluation Metrics

A. Experimental Setup and Hyperparameters

All architectures use an identical streaming pipeline with context window $C = 256$ and target horizon $S = 64$ ($W = C + S = 320$), trained at stride 16 and evaluated at stride 1. Context and target representations ($D = 32$) are extracted by the multiscale frequency front-end followed by a 6-block causal dilated TCN (64 filters, kernel size 5, dilations $\{1, 2, 4, 8, 16, 32\}$). Multi-statistic temporal pooling (terminal, mean, max, standard deviation, Log-Ricci energy, and first-order total variation) projects features through a linear bottleneck before LayerNorm and dropout ($p = 0.20$). Models are trained for 20 epochs on nominal sequences using AdamW (learning rate $10^{-3} \to 10^{-5}$ via cosine annealing, weight decay 10^{-4}, batch size 32), with identical training budgets and zero per-dataset hyperparameter tuning across families.

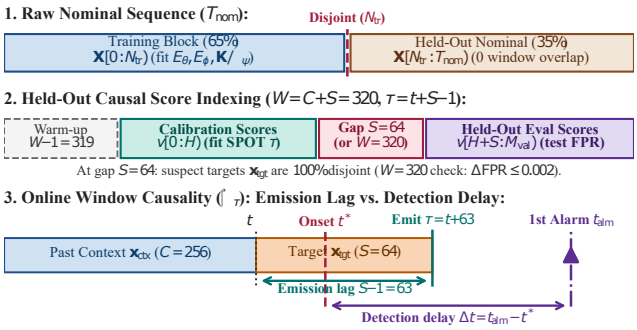


Fig. 2. Chronological split, boundary window indexing, and online causality protocol (C=256, S=64, W=320). Row 1 separates raw training (65%) and held-out nominal (35%) observations with zero window overlap across N_{tr}. Row 2 indexes causal held-out scores $v[0 : M_{\text{val}})$ into context warm-up ($W-1$=319), calibration $v[0 : H)$, isolation gap S=64 (or full-window W=320), and held-out evaluation $v[H+S : M_{\text{val}})$. Row 3 distinguishes structural score-emission lag ($S-1$=63) from empirical fault-detection delay ($\Delta t = t_{\text{alm}} - t^*$).

a) Held-Out Calibration Protocol and Boundary Isolation: To evaluate out-of-sample threshold calibration without circularity, each stream's raw nominal sequence $\mathbf{X}_{\text{nom}} \in \mathbb{R}^{T_{\text{nom}} \times K}$ is partitioned chronologically (Fig. 2, Row 1) into a *training block* $\mathbf{X}_{\text{tr}} = \mathbf{X}_{\text{nom}}[0 : N_{\text{tr}})$ ($N_{\text{tr}} = \lfloor 0.65\, T_{\text{nom}} \rfloor$) and a disjoint *held-out nominal block* $\mathbf{X}_{\text{val}} = \mathbf{X}_{\text{nom}}[N_{\text{tr}} : T_{\text{nom}})$. Standardization parameters and neural weights are fit on $\mathbf{X}_{\text{tr}}$ windows alone, guaranteeing zero raw-observation or window overlap across N_{tr}.

Within $\mathbf{X}_{\text{val}}$ ($N_{\text{val}} = T_{\text{nom}} - N_{\text{tr}}$), sliding windows of length $W = 320$ are constructed strictly inside $\mathbf{X}_{\text{val}}$ (Fig. 2, Row 2). Under causal trailing attribution, the first $W - 1 = 319$ observations provide context warm-up, yielding $M_{\text{val}} = N_{\text{val}} - W + 1$ valid score timestamps $v[0 : M_{\text{val}})$. Splitting v at midpoint $H = \lfloor M_{\text{val}}/2 \rfloor$ defines the *nominal calibration block* $v_{\text{cal}} = v[0 : H)$ (used to fit the SPOT threshold at $u = p_{98}, q = 10^{-3}$) and the *held-out evaluation block* $v_{\text{hold}} = v[H + S : M_{\text{val}})$, excluding $S = 64$ intermediate windows $v[H : H + S)$. Because suspect target windows have length $S = 64$, the last calibration target and first evaluation target are 100% disjoint. The first $C = 256$ evaluation steps share passive historical context with calibration observations, mirroring online streaming where past nominal history forms context for current predictions. To verify that shared passive context does not bias calibration, we also run a full-window isolation check ($G = W = 320$, excluding $v[H : H + 320)$ across the seven streams with $M_{\text{val}}/2 > 420$). Stream-macro causal FPR is virtually unchanged: 0.33% → 0.36% for `Soliton-Reynolds JEPA` and 3.02% → 3.24% for `TS-JEPA`, versus 12.15% → 13.97% for TimesNet and 9.05% → 9.73% for TranAD. All primary tables use fixed standard SPOT thresholds during testing.

b) Independent Statistical Units, Multi-Seed Replication, and Equivalence Bounds: Principal models on the

19 non-GHL streams are evaluated across five random seeds ($\{42, 123, 456, 7, 2024\}$); the 14 GHL streams ($> 10^6$ observations each) and ablation controls use three seeds ($\{7, 42, 2024\}$). To prevent pseudoreplication, every statistical test specifies its independent unit explicitly: (i) seeds are averaged within each stream first to quantify within-stream optimization variance ($\mathrm{SD}_{\mathrm{seed}}$), (ii) stream-level paired tests treat the telemetry stream ($n = 33$ primary or $n = 19$ non-GHL) as the unit of analysis, and (iii) domain-level tests aggregate streams within each hardware platform first and treat the dataset family ($N = 7$) as the independent cluster. We employ three statistical tools:

1) **Paired Wilcoxon signed-rank test and false-discovery-rate adjustment:** The Wilcoxon signed-rank test evaluates whether the median paired difference across streams ($n = 33$ or $n = 19$) is zero without Gaussian assumptions. We adjust p-values across multiple comparisons via the Benjamini–Hochberg false-discovery-rate (FDR) procedure ($q_{\mathrm{FDR}} = 0.05$), report 95% stream-resampled bootstrap confidence intervals (CIs; 2,000 resamples), and cross-check at the dataset-cluster level ($N = 7$) via exact sign-flip permutation tests and zero-tuning LODO sensitivity.
2) **Two One-Sided Tests (TOST) for fixed-backbone regularizer equivalence:** We use TOST to test whether the multiplicative soliton and manifold gain in (5) alters aggregate Point-F1 relative to the unmodulated multi-horizon whitened envelope $\mathcal{S}_{\mathrm{env}}(t)$ on the identical trained backbone ($n = 19$ streams). TOST tests $H_{01} : \mu_1 - \mu_2 \leq -\Delta$ (p_{lower}) and $H_{02} : \mu_1 - \mu_2 \geq +\Delta$ (p_{upper}), establishing equivalence when $p_{\mathrm{TOST}} = \max(p_{\mathrm{lower}}, p_{\mathrm{upper}}) < 0.05$ (meaning the 90% paired CI lies inside $[-\Delta, +\Delta]$) across $\Delta \in \{\pm 0.015, \pm 0.020, \pm 0.030\}$ (Table II(a)).
3) **Distributional tail and drift diagnostics (γ_2 and D_{KS}):** On nominal windows we measure: (i) *excess kurtosis* $\gamma_2 = \mathbb{E}[(s - \mu)^4]/\sigma^4 - 3$, distinguishing heavy-tailed *leptokurtic* ($\gamma_2 > 0$) from light-tailed *platykurtic* ($\gamma_2 < 0$) residuals; and (ii) the two-sample *Kolmogorov–Smirnov (KS) drift statistic* $D_{\mathrm{KS}} = \sup_x |F_{\mathrm{cal}}(x) - F_{\mathrm{hold}}(x)| \in [0, 1]$, quantifying distributional shift between calibration (F_{cal}) and held-out nominal (F_{hold}) blocks.

B. Baseline Architectures, Fairness Checks, and Cohort Structure

We separate primary multi-seed conclusions (Cohorts 1–2 across 12 neural, linear, and non-parametric arms) from secondary single-seed cross-checks. Two fairness audits verify that reconstructive baseline results under the unified 20-epoch, $W = 320$ pipeline are not artifacts of window length or implementation:

1) **Native published window lengths** (5 streams): Evaluating TranAD ($W = 10$) and TimesNet ($W = 96$) at their published window sizes raises Point-F1 (0.164 and 0.218), but observation residuals remain severely leptokurtic ($\gamma_2 =$ 448.3 and 98.1 vs. 3.1 for `TS-JEPA`) with high nominal FPR (28.4% and 44.1%).
2) **Secondary 130-stream single-seed cross-check:** Reconstructors replicate their published point-adjusted F1 on SMD and MSL, and Table III(d) provides a secondary 130-stream, 11-dataset single-seed comparison against NCAD, DCdetector, and Anomaly Transformer.

C. Cohort Summary and Reproducibility Specifications

Experiments are organized into five cohorts with distinct statistical roles:

- **Cohort 1 (33-stream primary multi-seed benchmark):** 19 non-GHL streams (5 seeds) and 14 GHL streams (3 seeds), reported in Tables III(a)–(c) and IV(a).
- **Cohort 2 (19-stream matched-backbone factorial and operating-curve suite):** 12 controlled arms across 5 seeds, reported in Table II(a),(c) and Fig. 4(b).
- **Cohort 3 (6-stream quiet-fault and hybrid-monitor suite):** Six injected single-channel fault types with hybrid sensor monitors (3 seeds), reported in Fig. 3.
- **Cohort 4 (5-stream horizon-latency and frequency-shell panels):** Target-horizon sweep ($S \in \{1, 4, 16, 32, 64\}$, 3 seeds, matched emission lag $S-1$) and dyadic front-end ablation, reported in Fig. 4(a) and Table II(b).
- **Cohort 5 (130-stream single-seed baseline screen):** Secondary breadth cross-check across 11 datasets against NCAD, DCdetector, and Anomaly Transformer (Table III(d)), treated strictly as auxiliary evidence rather than equivalent to the multi-seed benchmark.

At $K = 25$ channels, online trainable parameter counts are closely matched across latent and observation-space backbones: 272,266 online parameters for `TS-JEPA` (540,148 total including the stop-gradient EMA target encoder), 311,658 online (618,892 total including the multi-depth tap head and sub-horizon rollout predictor) for `Soliton-Reynolds JEPA`, 304,138 for matched-encoder `TCN-ObsPred`, 307,402 for `TCN-ObsPred-LN`, 648,553 for TimesNet, and 205,746 for TranAD. Non-finite telemetry entries are zero-imputed with channel standard deviations floored at 10^{-2}. Causal and retrospective buffered scores receive separately fitted thresholds τ_{SPOT} fitted strictly within their respective calibration windows v_{cal} as defined in (7). All primary multi-seed runs converged under the 20-epoch schedule on PyTorch 2.4 / CUDA 12.4 (NVIDIA RTX 5090, 0.60–0.88 ms per window, 188 MB of GPU memory); stream manifests, fitted thresholds, and source code are available at https://github.com/quaid281/NCAD.

D. Multi-Tier Statistical Evaluation Metrics

To guard against point-adjustment inflation, channel-averaging bias, and threshold gaming [10], [31], we evaluate models across three tiers. First, *primary calibration and accuracy outcomes* comprise realized held-out nominal FPR ($\mathrm{FPR}_{\mathrm{hold}}$), out-of-sample calibration error $|\mathrm{FPR}_{\mathrm{hold}} - q|$ relative to design risk $q = 10^{-3}$, unadjusted Point-F1, and threshold-free PR-AUC. Second, *operating characteristics* evaluate event recall across deployable calibration quantiles (q^*, Fig. 4(b)) and across post-hoc matched realized test-FPR budgets ($f^* \in \{0.01, 0.05\}$, Table II(c)). Third, *secondary operational endpoints* report unweighted Dataset-Macro Point-F1, event recall (fraction of contiguous anomaly segments with

≥ 1 alarm), mean detection delay Δt over detected events, and the constant-alarm baseline $\mathrm{F1}_{\text{all-pos}} = 2\pi/(1+\pi)$ at prevalence π.

VI. Ablation Studies and Analysis

A. Ablation 1: Multi-Horizon Whitened Envelope vs. Soliton Concentration Modulation

To separate the contribution of the multi-horizon barycenter-whitened envelope $\mathcal{S}_{\text{env}}(t)$ from the post-hoc Lieb–Thirring soliton multiplier on the same trained backbone, Table II(a) compares `Soliton-Reynolds JEPA` against its unmodulated control (`Soliton-Reynolds` (α_{sol}=0)) across all 19 non-GHL streams. Moving from single-horizon `TS-JEPA` to the unmodulated multi-horizon whitened envelope raises buffered PR-AUC from 0.331 to 0.374 (+0.043), cuts strictly causal held-out FPR six-fold (0.024 $\rightarrow$ 0.004), lowers calibration-to-holdout drift (D_{KS} : 0.214 $\rightarrow$ 0.162), and increases causal matched-FPR event recall at $f^* = 0.01$ from 0.395 to 0.697 (Table II(c)).

Adding the post-hoc Lieb–Thirring soliton concentration gain ($\alpha_{\text{sol}} = 0.12$) and nominal manifold gain on top of $\mathcal{S}_{\text{env}}(t)$ yields a null aggregate effect. Buffered held-out FPR shifts slightly from 0.122 to 0.108 (paired difference -0.013, 95% CI $[-0.035, +0.009]$), while causal FPR (0.004 vs. 0.003) and KS drift (0.162 vs. 0.163) are unchanged. Similarly, buffered SPOT Point-F1 (0.188 vs. 0.185, paired difference $+0.003$, Wilcoxon $p = 0.327$) and PR-AUC (0.375 vs. 0.374) satisfy TOST equivalence at $\Delta = \pm 0.030$ ($p_{\text{TOST}} = 0.0021$) and $\Delta = \pm 0.020$ ($p_{\text{TOST}} = 0.026$). All measurable gains over `TS-JEPA` in ranking, causal FPR, and drift suppression thus stem from multi-depth TCN tapping, barycenter centering, and multi-horizon Mahalanobis whitening rather than the post-hoc soliton multiplier.

B. Ablation 2: Fault-Class Breakdown, Quiet-Fault Mechanism, and Hybrid Monitor

Why do normalized latent encoders miss quiet single-channel faults such as sensor flatlines or dropouts? When a single channel k freezes or shifts by δ out of K channels, its contribution to the pre-LayerNorm pooled vector scales as $O(1/K)$ and is rescaled onto the $\sqrt{D}$ sphere by LayerNorm along with nominal cross-channel variance. Comparing the auxiliary context-variance input-channel gate against uniform channel weighting on four streams confirms that uniform weighting achieves higher macro Point-F1 (0.118 vs. 0.077), so the deployed pipeline retains uniform input encoding. Across six injected single-channel fault types on six multivariate streams (Fig. 3(b)), uniform latent encoding (`Unif.`) achieves 0.278 event recall on abrupt $+5\sigma$ spikes (0.306 with variance gating). On persistent quiet faults (flatlines, stuck sensors, dropouts) and low-amplitude steps, however, uniform latent encoding reaches only 0.194 event recall and 0.020–0.030 point recall (0.043–0.068 under variance or median absolute deviation (MAD) gating). Because standalone sensor monitors such as a rolling variance-collapse detector (`lowvar`) and maximum channel z-score (`dev_maxz`, 0.250–0.306 event recall at 14.1% FPR) trigger on different streams than `TS-JEPA`, combining them in a hybrid union detector (`TS-JEPA` $\vee$ `lowvar`) raises point recall to 0.202–0.318 and event recall to 0.417–0.500 across all six fault classes at a +15.1% nominal FPR cost (1.9% $\rightarrow$ 17.0%).

C. Ablation 3: Multiscale Dyadic-Shell Front-End

In Table II(b), replacing the standard convolutional stem with the 4-shell Fourier front-end increases macro PR-AUC from 0.213 to 0.263 (+0.050, led by SMD and GECCO) and lowers calibration residual kurtosis (1.409 $\rightarrow$ 1.069), while macro Point-F1 shifts from 0.118 to 0.078 due to conservative threshold tightening on SMAP `P-3`. The module thus acts as a ranking-conditioning stem rather than a uniform F1 gain.

D. Mechanism-Discrimination Study: Latent Space vs. Observation Space

Comparing the 12 factorial arms across all 19 non-GHL streams under identical splits, training budgets, and comparable parameter counts in Table II(c) isolates five architectural factors:

1) **Prediction vs. reconstruction in observation space (`TCN-ObsPred` vs. `TCN-ObsRecon`):** Using the exact same `HybridTCNEncoder` backbone, forecasting versus reconstructing raw observations leaves kurtosis ($\gamma_2 = 6.10$ vs. 6.24), calibration-to-holdout drift ($D_{\text{KS}} = 0.463$ vs. 0.445), and causal FPR (24.7% vs. 24.5%) virtually unchanged.
2) **LayerNorm bottleneck + saliency gate in observation space (`TCN-ObsPred-LN`):** Adding the $D = 32$ LayerNorm bottleneck and coordinate saliency gate to the observation forecaster (307k parameters) compresses tail kurtosis to $\gamma_2 = 0.91$ and halves causal FPR to 11.7%, but leaves calibration-to-holdout drift high ($D_{\text{KS}} = 0.376$) and lowers causal Point-F1 to 0.084.
3) **Learned latent target projection and multi-horizon barycenter whitening (`TS-JEPA` and `Soliton-Reynolds JEPA`):** Moving the prediction target into the learned $D = 32$ latent space halves calibration-to-holdout drift relative to `TCN-ObsPred-LN` (D_{KS} : 0.376 $\rightarrow$ 0.214 in `TS-JEPA` and 0.163 in `Soliton-Reynolds JEPA`, paired Wilcoxon $p \leq 0.001$), cuts causal FPR to 2.4% in `TS-JEPA` and 0.3% in `Soliton-Reynolds JEPA`, and raises buffered PR-AUC from 0.257 to 0.331 (`TS-JEPA`) and 0.375 (`Soliton-Reynolds JEPA`).
4) **Learned TCN encoder vs. linear subspace (`PCA-LinPred-LN`) and isolation forest:** `PCA-LinPred-LN` shares $D = 32$, LayerNorm, and the saliency gate, achieving 2.1% causal FPR via tail bounding, but its causal Point-F1 collapses to 0.017 (0.104 buffered; `IsoForest` event recall 0.149), proving that the learned nonlinear encoder is required for anomaly discriminability.
5) **Temporal foresight (`JEPA-CtxRecon`), EMA target (`JEPA-NoEMA`), and saliency gate (`TS-JEPA-NoGate`):** Autoencoding context embeddings (`JEPA-CtxRecon`)

TABLE II
ABLATION AND MECHANISM-DISCRIMINATION STUDIES: (A) FIXED-BACKBONE SOLITON MODULATION CONTROL AND TOST EQUIVALENCE (n=19, 5 SEEDS), (B) DYADIC FRONT-END (n=5), AND (C) 12-ARM MATCHED-BACKBONE FACTORIAL COMPARISON (n=19, 5 SEEDS). ER IS EVENT RECALL.

(a) Soliton Modulation Control on Fixed Backbone & TOST Equivalence

Model / Control	Pt-F1	PR	FPR	ER	ΔF1 [90%]	p_{TOST}
TS-JEPA (Unconstrained)	0.206 ± 0.025	0.331	0.093	0.477	Baseline	–
Soliton-Reynolds JEPA	0.188 ± 0.052	**0.375**	0.108	0.496	+0.003 [−.011, +.016]	0.0021*
$\alpha_{\text{sol}} = 0$ control	0.185 ± 0.053	**0.374**	0.122	0.515	–	(0.026†)
TimesNet (Baseline)	0.197 ± 0.018	0.329	0.300	**0.583**	–	–
TranAD (Baseline)	0.130 ± 0.004	0.287	0.307	0.490	–	–

*TOST equiv. between `Soliton-Reynolds JEPA` and its fixed-backbone α_{sol}=0 envelope control at $\Delta = \pm 0.030$ (p=0.0021); †equiv. at $\Delta = \pm 0.020$ (p=0.026; two-sided Wilcoxon p=0.327). Post-hoc soliton modulation is statistically null in Point-F1 and PR-AUC (buffered matched-FPR event recall at f^*=0.01 is 0.621 vs. 0.604; causal is 0.701 vs. 0.697 in Panel (c)).

(b) Dyadic-Shell Front-End On/Off

Stream (F1 / PR)	Std. Conv	Dyadic	ΔF1 / ΔPR
SMAP `P-3`	**0.301** / 0.280	0.115 / **0.296**	−0.186 / +0.017
MSL `M-1`	0.014 / **0.491**	**0.016** / 0.449	+0.002 / −0.043
SMD `machine-1-2`	0.000 / 0.028	**0.068** / **0.221**	+0.068 / +0.193
Daphnet `S01R01E1`	**0.179** / **0.103**	0.130 / 0.066	−0.048 / −0.037
GECCO `water_quality`	**0.097** / 0.161	0.062 / **0.282**	−0.035 / +0.121
Macro Mean	**0.118** / 0.213	0.078 / **0.263**	−0.040 / +0.050

(c) Controlled 12-Arm Mechanism-Discrimination Study Across 19 Non-GHL Streams (5 Seeds)

	Tail / Drift		Held-Out FPR		Point-F1		PR-AUC		Causal Event Recall			
Architecture Arm	γ_2	D_{KS}	Caus.	Buf.	Caus.	Buf.	Caus.	Buf.	SPOT	q^*=0.05	f^*=0.01	f^*=0.05
`TS-JEPA` (Unconstrained)	1.33	0.214	0.024	0.093	0.161	0.206	0.278	0.331	0.487	0.730	0.395	0.584
`Soliton-Reynolds JEPA`	2.31	**0.163**	**0.003**	0.108	0.060	0.188	**0.289**	**0.375**	0.506	0.809	**0.701**	0.809
`Soliton-Reynolds` (α_{sol}=0)	2.24	**0.162**	0.004	0.122	0.061	0.185	**0.288**	**0.374**	0.498	**0.820**	**0.697**	**0.820**
`JEPA-CtxRecon` (Context Autoenc.)	2.59	0.203	0.017	0.108	0.112	0.188	0.235	0.256	0.542	0.795	0.431	0.573
`JEPA-NoEMA` (Hard-Copy Target)	1.16	0.167	0.009	0.098	0.146	**0.235**	0.273	0.328	0.503	0.816	0.398	0.600
`TS-JEPA-NoGate` (Plain ℓ_2 Norm)	0.91	0.214	0.018	0.077	0.156	0.209	0.272	0.324	0.538	0.728	0.418	0.599
`PCA-LinPred-LN` (Linear Subspace)	0.95	0.233	0.021	0.186	0.017	0.104	0.210	0.244	0.452	0.665	0.377	0.624
`TCN-ObsPred-LN` (Obs. + LayerNorm)	0.91	0.376	0.117	0.193	0.084	0.126	0.235	0.257	0.421	0.665	0.329	0.541
`IsoForest` (Non-Parametric)	**0.70**	0.373	0.039	**0.056**	0.041	0.041	0.209	0.222	0.149	0.631	0.317	0.443
`TCN-ObsPred` (Unbounded Obs. Pred.)	6.10	0.463	0.247	0.313	0.140	0.145	0.272	0.310	0.546	0.629	0.171	0.383
`TCN-ObsRecon` (Unbounded Obs. Recon.)	6.24	0.445	0.245	0.304	0.134	0.135	0.261	0.301	0.528	0.619	0.163	0.379
`TimesNet` (Baseline)	5.54	0.438	0.243	0.300	**0.209**	0.197	0.281	0.329	**0.588**	0.618	0.217	0.407
`TranAD` (Baseline)	5.64	0.459	0.222	0.307	0.140	0.130	0.248	0.287	0.533	0.602	0.149	0.343

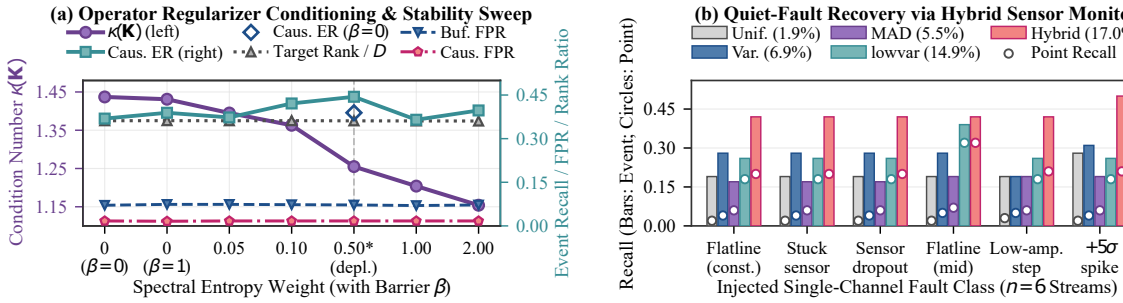


Fig. 3. Latent representation rank conditioning and quiet-fault recovery (n=6 streams, three seeds). **(a)** Operator condition number κ, target effective rank ratio $\text{RankMe}(z_{\text{tgt}})/D$, event recall (ER), and false-positive rate (FPR) across spectral regularization strengths. **(b)** Injected single-channel event recall (bars) and point recall (circles) across uniform TS-JEPA (`Unif.`), context variance (`Var.`) and median absolute deviation (`MAD`) gates, standalone rolling variance monitor (`lowvar`), and the hybrid union monitor (`TS-JEPA ∨ lowvar`; nominal FPR in parentheses).

reduces drift (D_{KS} = 0.203, causal FPR 1.7%) at lower causal F1 (0.112 vs. 0.161); hard-copy targets (`JEPA-NoEMA`) reach 0.9% causal FPR and 0.146 causal F1; and replacing the saliency gate with plain ℓ_2 norm (`TS-JEPA-NoGate`) yields 1.8% causal FPR and 0.156 causal F1, confirming that false-alarm reduction does not depend on the saliency gate.

VII. RESULTS AND DISCUSSION

A. Benchmark Performance on Continuous Telemetry

Table III reports all 33 continuous telemetry streams under the chronological held-out SPOT protocol ($u = p_{98}, q = 10^{-3}$): Panel (a) uses retrospective buffered attribution, Panel (b) strictly causal streaming, and Panel (c) per-dataset breakdowns including anomaly event counts (E) and both causal and buffered held-out nominal FPRs. Evaluating realized FPR and out-of-sample calibration error $|\text{FPR}_{\text{hold}} - q|$ as primary outcomes shows a large relative reduction in false alarms and calibration error, while `Soliton-Reynolds JEPA` achieves the highest buffered PR-AUC (0.375) and causal matched-FPR event recall (0.701 at $f^* = 0.01$).

Lower Realized FPR and Calibration Error on Held-Out Telemetry: On the 19 non-GHL streams (Table III(a),(b)), latent scoring cuts held-out nominal FPR and calibration error $|\text{FPR}_{\text{hold}} - q|$ three-fold under buffered attribution: 9.3% for `TS-JEPA` and 10.8% for `Soliton-Reynolds JEPA`, versus 30.0% for TimesNet ($p = 0.049$) and 30.7% for TranAD ($p = 0.007$). Under strictly causal streaming, the false-alarm reduction widens to ten- to eighty-fold: 2.4% for `TS-JEPA` and 0.3% for `Soliton-Reynolds JEPA`, compared to 24.3% for TimesNet ($p = 0.010$) and 22.2% for TranAD ($p = 0.010$). This advantage holds across all 33 streams (0.25% causal stream-macro FPR for `Soliton-Reynolds JEPA` and 1.51% for `TS-JEPA` vs. 16.11% and 14.23% for TimesNet and TranAD, Table IV(a)) and across the $N = 7$ dataset families ($p = 0.016$). Under zero-tuning LODO evaluation, leaving one dataset out at a time keeps causal FPR at 0.2%–2.3% (3.8%–7.0% buffered) for JEPAs versus 12.1%–18.4% causal (8.4%–22.1% buffered) for reconstructors. In Panel (c), `Soliton-Reynolds JEPA` achieves the lowest causal held-out FPR on every multi-stream family (0.2% Daphnet, 0.1% GHL, 0.4% MSL, 0.6% SMAP vs.

TABLE III
HELD-OUT CALIBRATION BENCHMARK (SPOT, $u{=}p_{98}$, $q{=}10^{-3}$): (A) RETROSPECTIVE BUFFERED ($S{=}64$) AND (B) STRICTLY CAUSAL TRAILING ATTRIBUTION ACROSS 19 NON-GHL STREAMS AND 14 GHL STREAMS, (C) PER-DATASET BREAKDOWN (INCLUDING ANOMALY EVENT COUNTS E AND CAUSAL/BUFFERED HELD-OUT FPR) ACROSS $N{=}7$ TELEMETRY FAMILIES, AND (D) 130-STREAM SINGLE-SEED CROSS-CHECK. BEST IN **BOLD**, SECOND UNDERLINED.

	(a) Retrospective Buffered, Non-GHL (19 Streams)					(b) Strictly Causal Trailing, Non-GHL (19 Streams)				GHL Buffered (14)		Macro
Architecture	**FPR ↓**	**Point-F1 ↑**	**PR-AUC ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Point-F1 ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Point-F1 ↑**	**DS-F1 ↑**
Latent-space predictive architectures (JEPAs)												
TS-JEPA (unconstrained)	0.093	**0.206**	0.331	0.477	110.3	0.024	0.161	0.487	124.0	0.004	0.000	**0.115**
Soliton-Reynolds JEPA	0.108	0.188	**0.375**	0.496	231.8	**0.003**	0.060	0.506	166.2	**0.002**	0.000	0.098
Soliton-Reynolds ($\alpha_{\text{sol}}{=}0$)	0.122	0.185	**0.374**	0.515	215.0	0.004	0.061	0.498	138.3	**0.002**	0.000	0.097
Observation-space reconstructive baselines												
TimesNet	0.300	0.197	0.329	**0.583**	**48.5**	0.243	**0.209**	**0.588**	**62.2**	0.061	**0.022**	0.100
TranAD	0.307	0.130	0.287	0.490	234.4	0.222	0.140	0.533	241.0	0.046	0.020	0.069
Non-neural linear baseline												
Linear AR Forecaster	**0.005**	0.051	0.255	0.432	87.8	–	–	–	–	**0.002**	0.000	0.038

	(c) Telemetry Family Descriptors					Buffered Point-F1 (Stream Mean) ↑				Causal / Buffered Held-Out Nominal FPR ↓			
Dataset	n	K	E	π	$\text{F1}_{\text{all-pos}}$	**TS-JEPA**	**Sol.-Rey.**	**TimesNet**	**TranAD**	**TS-JEPA**	**Sol.-Rey.**	**TimesNet**	**TranAD**
Daphnet	6	9	72	0.148	0.256	0.145	0.125	**0.229**	0.144	.030/.091	**.002/.038**	.291/.313	.250/.298
GHL	14	16	26	0.013	0.026	0.000	0.000	**0.022**	0.020	.004/.004	**.001/.002**	.056/.061	.040/.046
Genesis	1	18	2	0.003	0.006	0.000	**0.037**	0.000	0.000	.004/.017	.003/.073	**.000/.000**	**.000/.000**
MSL	4	55	6	0.255	0.381	**0.161**	0.138	0.158	0.151	.054/**.095**	**.004**/.122	.161/.133	.132/.241
SMAP	6	25	7	0.150	0.236	**0.379**	0.370	0.289	0.167	.014/**.135**	**.006**/.212	.333/.549	.360/.513
Metro	1	5	25	0.001	0.003	0.000	0.000	**0.001**	0.000	.008/.014	.001/**.000**	**.000/.000**	**.000/.000**
SWAN	1	38	6615	0.279	0.437	**0.119**	0.017	0.000	0.000	**.000/.000**	**.000/.000**	**.000/.000**	**.000/.000**

Note: Calibration error $|\text{FPR}_{\text{hold}} - q|$ at $q{=}0.001$ is FPR $- 0.001$. 95% stream-bootstrap CIs (19 streams) for (a) Buffered FPR/F1: TS-JEPA [.048, .145]/[.113, .319], Sol.-Rey. [.047, .179]/[.093, .291], ($\alpha_{\text{sol}}{=}0$) [.060, .195]/[.091, .289], TimesNet [.162, .446]/[.098, .315], TranAD [.177, .445]/[.053, .232], Linear AR [.001, .011]/[.012, .104]; (b) Causal FPR/F1: TS-JEPA [.007, .046]/[.063, .287], Sol.-Rey. [.002, .004]/[.014, .119], ($\alpha_{\text{sol}}{=}0$) [.002, .005]/[.014, .120], TimesNet [.123, .375]/[.109, .326], TranAD [.119, .337]/[.058, .241]. In (b), causal test alert rate $\hat{p}_{\text{alert}}$/TPR/rate-matched random-alerter $\text{ER}_{\text{rand}}(p{=}\text{FPR}_{\text{hold}})$ ($\text{ER}_{\text{rand}}(p{=}\hat{p}_{\text{alert}})$) across 19 streams (111 events on 18 non-SWAN streams, median $L{=}139$, mean 348.1; 6,615 SWAN events): TS-JEPA .108/.166/.283 (.656), Sol.-Rey. .022/.052/.288 (.524, $\Delta t_{\text{rand}}{=}214.6$ at FPR_{hold}, 114.7 at $\hat{p}_{\text{alert}}$), ($\alpha_{\text{sol}}{=}0$) .021/.052/.288 (.509), TimesNet .242/.340/.600 (.801), TranAD .213/.239/.601 (.801). In (c), E is test events (34, 3, 2, 5, 7, 13, 3,137 in held-out half); Causal/Buffered FPR stream SDs: Daphnet .001–.030/.032–.091 vs. .337–.402/.381–.430 (seed SD .002–.041); GHL .001–.005/.002–.006 vs. .041–.065; MSL .001–.047/.112–.147 vs. .228–.281; SMAP .003–.019/.125–.181 vs. .080–.150. **(d) 130-Stream Single-Seed Cross-Check (Point-F1, dataset-macro / stream mean, 11 datasets):** NCAD 0.058/0.088; DCdetector 0.070/0.097; Anomaly Transformer 0.091/0.082; TimesNet 0.129/0.151; TranAD 0.156/0.152; TS-JEPA **0.160 / 0.186**.

5.6%–36.0% for baselines).

Multi-Horizon Ranking Gains vs. Unadjusted Point-F1 Trade-offs: Under buffered attribution on the 19 non-GHL streams (Table III(a)), `Soliton-Reynolds JEPA` and its $\alpha_{\text{sol}} = 0$ control achieve the highest threshold-free PR-AUC (0.375 and 0.374 vs. 0.331 for `TS-JEPA`, 0.329 for TimesNet, and 0.287 for TranAD). Buffered Point-F1 (0.188 and 0.185, vs. 0.206 for `TS-JEPA`) and event recall (0.496 and 0.515) match TimesNet (0.197 F1, 0.583 recall) and beat TranAD (0.130 F1, 0.490 recall) at one-third the false-alarm rate. Across datasets (Table III(c)), SMAP exceeds the trivial always-alarm F1 of 0.236 (0.379 and 0.370; PR-AUC 0.576 vs. $\pi = 0.150$) and Genesis reaches 0.037 F1 (vs. 0.000). Conversely, on GHL ($\pi = 0.013$), all JEPAs and the linear autoregressive (AR) forecaster fire near-silently (0.1%–0.4% FPR, 0.000 F1), illustrating that low FPR without recall does not indicate detection.

Domain and Operational Applicability Boundaries: Synthesizing performance across fault classes and sampling rates defines four operational regimes:

1) *Favorable operating regime:* Multi-channel spacecraft and wearable kinematic arrays ($K \geq 9$, SMAP, MSL, Daphnet) with persistent localized transients, where suppressing nuisance false alarms under automated calibration is the primary operational requirement.
2) *Latency-constrained telemetry:* Systems where a 63-step emission lag is prohibitive for immediate tripwire response (such as hourly Metro where 63 steps = 63 h), requiring short horizons $S \in \{1, 4\}$ (Section VII-D).
3) *Quiet single-channel faults:* Inactive sensor modes (flatlines, dropouts) that LayerNorm embeddings absorb as nominal background, requiring the hybrid union detector (`TS-JEPA ∨ lowvar`) when single-channel coverage is mandatory (Section VI-B).
4) *Continuous non-stationary process ramps:* Industrial heating loops and traffic drifts (GHL, Metro; 0.000–0.022 F1), which defeat both static and rolling thresholds.

B. Component-Level Analysis of Soliton-Reynolds JEPA

Tables II(a),(c) and III(a),(b) isolate how each component of `Soliton-Reynolds JEPA` modifies performance relative to `TS-JEPA`. First, *multi-depth TCN tapping and barycenter centering* remove common-mode baseline shift, cutting KS drift ($D_{\text{KS}} : 0.214 \rightarrow 0.163$) and eliminating the inter-block drift share of excess causal FPR (71.3% → 0.0%, Table IV(b)). Second, *nested multi-horizon rollout and Mahalanobis whitening* ($\mathcal{S}_{\text{env}}(t)$, $\alpha_{\text{sol}} = 0$) capture sub-horizon transients across $S_m \in \{16, 32, 48, 64\}$, raising buffered PR-AUC (0.331 → 0.374), cutting causal FPR (2.4% → 0.4%), and lifting causal matched-FPR event recall at $f^* = 0.01$ (0.395 → 0.697). Third, *post-hoc Lieb–Thirring soliton modulation* ($\alpha_{\text{sol}} = 0.12$) and manifold gain yield a null aggregate effect on the fixed backbone (0.188 vs. 0.185 F1, $p_{\text{TOST}} = 0.0021$, Wilcoxon $p = 0.327$; 0.375 vs. 0.374 PR-AUC), confirming that the multi-horizon whitened representation drives the gains.

C. Operational Incident Evaluation and Causal-vs-Buffered Trade-offs

Comparing Panels (a) and (b) of Table III shows how zero-lookahead streaming shifts the operating threshold. In

causal mode, Soliton-Reynolds JEPA preserves the highest threshold-free ranking (PR-AUC 0.289 vs. 0.278 for TS-JEPA and 0.281 for TimesNet). Without 64-step window averaging, however, causal 128-dimensional whitened residuals have a heavier right tail (median $\xi = -0.182$, $\gamma_2 = 2.31$) than buffered scores ($\xi = -0.473$, $\gamma_2 = 1.58$) or single-horizon TS-JEPA ($\xi = -0.307$, $\gamma_2 = 1.33$). When SPOT extrapolates that tail to $q = 10^{-3}$ in (7), it places τ_{SPOT} high on the causal score distribution. Consequently, Soliton-Reynolds JEPA fires quietly on only $\hat{p}_{\text{alert}} = 2.2\%$ of test steps (true-positive rate (TPR) 5.2%, held-out FPR 0.3%), compared to 10.8% for TS-JEPA (TPR 16.6%, held-out FPR 2.4%) and 24.2% for TimesNet (TPR 34.0%, held-out FPR 24.3%). Because it flags only the sharpest peak steps inside each anomaly segment rather than staying on throughout the event, its causal Point-F1 drops (0.060 vs. 0.188 buffered, and 0.161 for TS-JEPA) even though it catches 50.6% of anomaly events ($\Delta t = 166.2$).

To check whether 50.6% event recall could arise by chance on long anomaly segments (111 events with median length $L = 139$ and mean 348.1 across the 18 non-SWAN streams), Table III (Note) compares each detector against an analytical rate-matched i.i.d. Bernoulli random alerter, $\text{ER}_{\text{rand}}(p) = E^{-1}\sum_{e=1}^{E}[1-(1-p)^{L_e}]$. At Soliton-Reynolds JEPA's held-out nominal rate ($p = \text{FPR}_{\text{hold}} = 0.0033$), random firing catches only $\text{ER}_{\text{rand}} = 0.288$ of events ($\Delta t_{\text{rand}} = 214.6$ vs. 0.506 and 166.2 actual). At its overall test firing rate ($\hat{p}_{\text{alert}} = 0.0220$), unclustered Bernoulli trials reach $\text{ER}_{\text{rand}} = 0.524$ ($\Delta t_{\text{rand}} = 114.7$) because real model alarms cluster inside detected events rather than firing independently. For TimesNet ($p_{\text{hold}} = 0.243$), independent random firing reaches 0.600 (0.801 at $\hat{p}_{\text{alert}}$), exceeding TimesNet's actual 0.588. Relaxing the static threshold to the empirical $p_{99.5}$ calibration quantile raises Soliton-Reynolds JEPA's causal event recall to 0.612 at 0.76% held-out FPR.

D. Horizon Length, Score-Emission Latency, and False-Alarm Control

Target horizon $S = 64$ imposes a 63-step emission lag before an alarm can trigger (Section IV-C). To evaluate detection quality versus latency against baselines at matched emission lag $S - 1 \in \{0, 3, 15, 31, 63\}$, Fig. 4(a) retrains models across $S \in \{1, 4, 16, 32, 64\}$ on five streams (three seeds). At $S = 1$ (zero lag) and $S = 4$ (3-step lag), TS-JEPA holds causal FPR to 0.4% and 0.6% (vs. 2.5% and 3.1% for TimesNet) while achieving higher causal PR-AUC (0.178–0.189 vs. 0.150–0.162 for TimesNet), comparable Point-F1 at $S = 1$ (0.028 vs. 0.025), and fast detection ($\Delta t = 57.8$ and 85.2 steps). As S increases across $\{16, 32, 64\}$, TimesNet causal FPR escalates nearly nine-fold (6.1% → 9.8% → 21.3%), whereas latent predictors remain bounded ($\leq 2.0\%$). Consequently, $S = 64$ suits high-rate sensors (≥ 1 Hz, such as 64 Hz Daphnet where 63 steps = 0.98 s) or slow subsystem trend monitoring (0.1 Hz SMAP, MSL, and GHL, 10.5 min). Low-rate infrastructure streams ($\leq 1\,\text{min}^{-1}$, such as hourly Metro where 63 h lag is prohibitive) and rapid tripwires instead require $S \in \{1, 4\}$.

TABLE IV
THRESHOLD CALIBRATION ANALYSIS AT DESIGN RISK q=10^{-3}: (A) PARAMETRIC SPOT VS. NON-PARAMETRIC 99.5TH PERCENTILE ACROSS 33 STREAMS (CELL-WEIGHTED MEANS, STREAM-MACRO IN PARENTHESES, AND CAUSAL CALIBRATION ERROR $|\text{FPR}_{\text{HOLD}} - q|$), AND (B) THREE-STAGE REFITTED PERMUTATION DECOMPOSITION OF EXCESS CAUSAL FPR ACROSS ALL 19 NON-GHL STREAMS.

(a) Model	Rule	Causal FPR	$\lvert\text{FPR}-q\rvert$ (Over.)	Buffered FPR	F1 (buf)
TS-JEPA	Standard SPOT	0.017 (0.015)	.014 (15×)	0.064 (0.055)	0.139
	Empirical p99.5	0.025 (0.023)	.022 (23×)	0.076 (0.066)	0.140
Sol.-Rey.	Standard SPOT	**0.003 (0.003)**	**.002 (3×)**	0.076 (0.063)	0.130
	Empirical p99.5	0.007 (0.007)	.006 (7×)	0.091 (0.077)	0.138
TimesNet	Standard SPOT	0.180 (0.161)	.160 (161×)	0.222 (0.199)	0.140
	Empirical p99.5	0.191 (0.175)	.174 (175×)	0.234 (0.213)	0.141
TranAD	Standard SPOT	0.160 (0.142)	.141 (142×)	0.222 (0.196)	0.094
	Empirical p99.5	0.171 (0.154)	.153 (154×)	0.234 (0.209)	0.096

	Refitted Stage Causal FPR			Share of Excess (%)		
(b) Model	FPR_{chr}	FPR_{blk}	FPR_{pt}	Inter-Blk	Serial	i.i.d.
TS-JEPA	0.0229	0.0073	0.0033	71.3	18.1	10.5
Soliton-Rey.	**0.0033**	**0.0037**	**0.0025**	**0.0**	44.8	55.2
TimesNet	0.2302	0.0427	0.0035	81.8	17.1	1.1
TranAD	0.2103	0.0464	0.0034	78.3	20.5	1.1

In (a), Over. is the causal overshoot ratio $\text{FPR}_{\text{macro}}/q$ at q=10^{-3}; under S=64 buffered attribution, window smearing dilates single exceedances across 64 steps (55–213 × q). In (b), 95% stream-bootstrap CIs for inter-block drift $\text{FPR}_{\text{chr}} - \text{FPR}_{\text{blk}}$ (n=19): TS-JEPA $+.0156$ $[+.001, +.034]$, Soliton-Rey. $-.0004$ $[-.002, +.001]$, TimesNet $+.1874$ $[+.078, +.305]$, TranAD $+.1639$ $[+.072, +.267]$.

E. Deployable Calibration Quantiles vs. Matched Realized Test-FPR Operating Points

Because reconstruction baselines calibrate to 22%–31% realized FPR versus 0.3%–11% for latent predictors, and long segments (402 steps on Daphnet, 1,102 on SMAP) allow a 30% random-alarm detector to touch > 95% of segments via $1 - (1 - \text{FPR})^L$, we compare models under two operating regimes:

1) **Deployable nominal calibration quantiles (q^*, Fig. 4(b), Table II(c)):** At causal calibration quantile $q^* = 0.050$ (set without test labels), TS-JEPA and Soliton-Reynolds JEPA achieve higher event recall (0.730 and 0.809) at lower false-alarm rates than TimesNet (0.618 recall) and TranAD (0.602 recall).
2) **Post-hoc matched realized test-FPR budgets (f^*, Table II(c)):** Equalizing realized test nominal FPR across models, at $f^* = 0.01$ TS-JEPA (0.395) and Soliton-Reynolds JEPA (0.701 causal, 0.621 buffered) substantially increase event recall compared to TimesNet (0.217) and TranAD (0.149), whereas at relaxed budgets ($f^* \geq 0.10$) recall curves converge ($p = 0.625$). Latent scoring thus specifically improves sensitivity in the low-FPR regime ($f^* \leq 0.05$).

F. Automated Threshold Calibration Realities on Non-Stationary Telemetry

Static extreme-value thresholding does not achieve its nominal design risk $q = 10^{-3}$ (0.1%) out of sample across all streams and attribution modes. Across all 33 streams (Table IV(a)), standard SPOT achieves stream-macro causal FPR of 1.51% for TS-JEPA ($15 \times q$) and 0.25% for Soliton-Reynolds JEPA ($3 \times q$), compared to 16.11% for TimesNet ($161 \times q$) and 14.23% for TranAD ($142 \times q$). Replacing parametric SPOT with a non-parametric empirical 99.5th percentile ($p_{99.5}$) yields nearly identical results (0.7% causal FPR for Soliton-Reynolds JEPA and 2.3% for TS-JEPA vs. 15.4%–17.5% for reconstructors), demonstrating that reconstructor false-alarm flooding is not an artifact

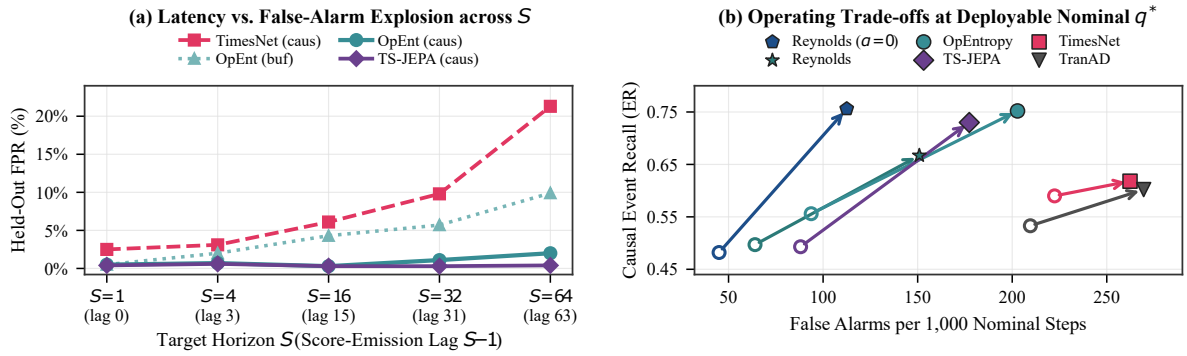


Fig. 4. Horizon length and deployable calibration trade-offs. **(a)** Held-out FPR and event recall across target horizons $S \in \{1, 4, 16, 32, 64\}$ at matched score-emission latency $S-1 \in \{0, 3, 15, 31, 63\}$ steps (five streams, three seeds). **(b)** Causal event recall versus false alarms across 19 non-GHL streams as nominal calibration quantiles shift from $q^*{=}10^{-3}$ (open circles) to $q^*{=}0.05$ (filled markers).

of GPD tail fitting. Under $S = 64$ buffered attribution, however, smearing each window score across 64 consecutive steps dilates every isolated exceedance into a 64-step cluster, inflating buffered FPR to 5.5%–6.3% (55–63 $\times$ q) for JEPAs and 19.6%–19.9% (196–199 $\times$ q) for reconstructors under both rules. Static SPOT is therefore near-calibrated for `Soliton-Reynolds JEPA` only in strictly causal mode. Two diagnostics explain why causal thresholds hold on latent residuals:

1) **Three-stage refitted permutation decomposition (Table IV(b)):** To separate inter-block regime drift from serial correlation and GPD tail-fit error, we pool nominal calibration and evaluation scores $\mathbf{v}_{\text{pool}} = [v_{\text{cal}}, v_{\text{hold}}]$ across all 19 non-GHL streams and *refit* SPOT ($u = p_{98}, q = 10^{-3}$) under three splits: chronological (FPR_{chr}), $L = 64$ block permutation (FPR_{blk}, which removes macro-block drift while preserving intra-block serial correlation), and pointwise i.i.d. permutation (FPR_{pt}). Once temporal structure is removed by i.i.d. shuffling, every model achieves low false-alarm rates near the 0.1% target ($\text{FPR}_{\text{pt}} = 0.0025$ for `Soliton-Reynolds JEPA` and 0.0033–0.0035 for `TS-JEPA`, TimesNet, and TranAD). In chronological deployment, inter-block drift ($\text{FPR}_{\text{chr}} - \text{FPR}_{\text{blk}}$) accounts for 71.3% of excess causal FPR in single-horizon `TS-JEPA` (0.0229 $\rightarrow$ 0.0073) and 78.3%–81.8% in reconstructors (0.2103–0.2302 $\rightarrow$ 0.0427–0.0464). In barycenter-centered `Soliton-Reynolds JEPA`, by contrast, inter-block drift vanishes (0.0%, $\text{FPR}_{\text{chr}} = 0.0033$ vs. $\text{FPR}_{\text{blk}} = 0.0037$, difference -0.0004).
2) **GPD tail stability, declustering, and rolling adaptation:** On the 19 non-GHL streams at $u = p_{98}$, median calibration exceedance count is $N_u = 28$ causally (21 buffered) with 0% solver fallback and low threshold uncertainty ($\text{SE}(\tau)/\tau = 2.6\%$–$3.5\%$). Grimshaw GPD fits hit the lower shape boundary ($\xi = -0.90$) in only 4.4% of JEPA cells (median $\xi = -0.27$) versus 34.4% for TimesNet and 54.4% for TranAD. Applying Ferro–Segers runs declustering ($r = 5$) [30] prior to GPD fitting trims stationary causal FPR from 0.25% to 0.20% for `Soliton-Reynolds JEPA` and from 0.72% to 0.44% for `TS-JEPA`. Under discrete step, variance, and regime shifts across 16 streams, adaptive rolling quantiles hold causal `TS-JEPA` FPR to 0.06%–0.40% (vs. 0.8%–3.0% for TimesNet), whereas continuous monotonic mean ramps defeat trailing rolling windows (32.6%–38.5% FPR across all models).

VIII. Conclusion and Future Work

We evaluated how Joint-Embedding Predictive Architectures condition anomaly residuals under automated extreme-value calibration across 33 continuous telemetry streams under a chronological held-out protocol:

1) **Factorial Mechanism Isolation and Multi-Horizon Whitening:** Matched-encoder ablations (272k–312k online parameters) decouple score compression from representation learning. LayerNorm bottlenecks bound residual kurtosis (γ_2 : 6.10 $\rightarrow$ 0.91), learned latent projections halve calibration-to-holdout drift (D_{KS} : 0.376 $\rightarrow$ 0.214 in `TS-JEPA`), and multi-horizon barycenter whitening in `Soliton-Reynolds JEPA` lowers drift further to $D_{\text{KS}} = 0.163$ with 0.3% causal FPR. On the fixed backbone, post-hoc Lieb–Thirring soliton modulation is a null modification satisfying TOST equivalence ($\Delta = \pm 0.020$, $p_{\text{TOST}} = 0.026$, Wilcoxon $p = 0.327$), and short horizons ($S \in \{1, 4\}$) remove the 63-step emission lag.
2) **Relative False-Alarm Control, Threshold Trade-offs, and Hybrid Quiet-Fault Recovery:** Latent scoring lowers held-out false alarms three-fold under buffered attribution and ten- to eighty-fold causally. `Soliton-Reynolds JEPA` achieves leading buffered PR-AUC (0.375) and causal matched-FPR event recall ($f^* = 0.01$: 0.701 vs. 0.217 for TimesNet), trading lower causal SPOT Point-F1 (0.060 at a 2.2% test alert rate vs. 0.161 at 10.8% for `TS-JEPA`) for 0.3% held-out FPR. Pairing `TS-JEPA` with a rolling variance monitor raises quiet single-channel event recall from 0.19–0.28 to 0.42–0.50 at a +15.1% nominal FPR cost.
3) **Calibration Drift and Adaptive Thresholding:** Static SPOT overshoots $q = 10^{-3}$ by 55–126$\times$ under $S = 64$ buffered window smearing and 15$\times$ causally for `TS-JEPA`. In contrast, barycenter-whitened `Soliton-Reynolds JEPA` reduces causal static overshoot to 3$\times$ (0.25% stream-macro FPR across 33 streams; $\text{FPR}_{\text{pt}} = 0.0025$) and eliminates inter-block drift (71.3% $\rightarrow$ 0.0%). Runs-declustered SPOT and rolling quantiles narrow this gap across step and

variance shifts, motivating regime-conditioned conformal calibration.

REFERENCES

[1] F. Meng, B. Yuan, and F. Gao, "Spacecraft telemetry anomaly detection based on deep learning: A review and benchmark," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 56, no. 5, pp. 3863–3878, 2020.

[2] R. Chalapathy and S. Chawla, "Deep learning for anomaly detection: A survey," *arXiv preprint arXiv:1901.03407*, 2019.

[3] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, and T. Soderstrom, "Detecting spacecraft anomalies using lstms and nonparametric dynamic thresholding," *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pp. 387–395, 2018.

[4] J. Xu, H. Wu, J. Wang, and M. Long, "Anomaly transformer: Time series anomaly detection with association discrepancy," in *International Conference on Learning Representations (ICLR)*, 2022.

[5] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, and M. Long, "Timesnet: Temporal 2d-variation modeling for general time series analysis," in *International Conference on Learning Representations (ICLR)*, 2023.

[6] S. Tuli, G. Casale, and N. R. Jennings, "Tranad: Deep transformer networks for anomaly detection in multivariate time series data," *Proceedings of the VLDB Endowment*, vol. 15, no. 6, pp. 1201–1214, 2022.

[7] A. Siffer, P.-A. Fouque, A. Termier, and C. Largouet, "Anomaly detection in streams with extreme value theory," in *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2017, pp. 1067–1075.

[8] Y. LeCun, "A path towards autonomous machine intelligence," *OpenReview Preprint*, 2022.

[9] C. U. Carmona, F.-X. Aubet, V. Flunkert, and J. Gasthaus, "Neural contextual anomaly detection for time series," in *International Conference on Learning Representations (ICLR)*, 2022.

[10] S. Schmidl, P. Wenig, and T. Papenbrock, "Anomaly detection in time series: a comprehensive evaluation," *Proceedings of the VLDB Endowment*, vol. 15, no. 9, pp. 1779–1797, 2022.

[11] Z. Yue, Y. Wang, J. Duan, T. Yang, C. Huang, Y. Tong, and B. Xu, "Ts2vec: Towards universal representation of time series," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 8, 2022, pp. 8980–8987.

[12] Y. Yang, C. Zhang, T. Zhou, Q. Gao, Q. Wen, and Y. Liang, "Dcdetector: Dual attention contrastive representation learning for time series anomaly detection," in *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023, pp. 3033–3045.

[13] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, and N. Ballas, "Self-supervised learning from images with a joint-embedding predictive architecture," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 15 619–15 629.

[14] A. Bardes, Q. Duval, C. Feichtenhofer, I. Laptev, D. Haziza, V. Khalidov, N. Ballas, and Y. LeCun, "V-jepa: Video joint-embedding predictive architecture," *arXiv preprint arXiv:2404.08471*, 2024.

[15] J.-B. Grill, F. Strub, F. Altché, C. Tallec, P. Richemond, E. Buchatskaya, C. Doersch, B. Avila Pires, Z. Guo, M. Gheshlaghi Azar *et al.*, "Bootstrap your own latent-a new approach to self-supervised learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 21 271–21 284.

[16] A. Bardes, J. Ponce, and Y. LeCun, "Vicreg: Variance-invariance-covariance regularization for self-supervised learning," in *International Conference on Learning Representations (ICLR)*, 2022.

[17] J. Zbontar, L. Jing, I. Misra, Y. LeCun, and S. Deny, "Barlow twins: Self-supervised learning via redundancy reduction," in *International Conference on Machine Learning (ICML)*, 2021, pp. 12 310–12 320.

[18] A. Ermolov, A. Siarohin, E. Sangineto, and N. Sebe, "Whitening for self-supervised representation learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021, pp. 8315–8327.

[19] S. L. Brunton, J. L. Proctor, and J. N. Kutz, "Discovering governing equations from data by sparse identification of nonlinear dynamical systems," *Proceedings of the National Academy of Sciences*, vol. 113, no. 15, pp. 3932–3937, 2016.

[20] R. T. Chen, Y. Rubanova, J. Bettencourt, and D. K. Duvenaud, "Neural ordinary differential equations," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31, 2018, pp. 6571–6583.

[21] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.

[22] S. Greydanus, M. Dzamba, and J. Yosinski, "Hamiltonian neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 15 379–15 389.

[23] M. Cranmer, S. Greydanus, S. Hoyer, P. Battaglia, D. Spergel, and S. Ho, "Lagrangian neural networks," *arXiv preprint arXiv:2003.04630*, 2020.

[24] S. Wang, Y. Teng, and P. Perdikaris, "Understanding and mitigating gradient pathologies in physics-informed neural networks," *SIAM Journal on Scientific Computing*, vol. 43, no. 5, pp. A3055–A3081, 2021.

[25] B. O. Koopman, "Hamiltonian systems and transformations in Hilbert space," *Proceedings of the National Academy of Sciences*, vol. 17, no. 5, pp. 315–318, 1931.

[26] I. Mezić, "Spectral properties of dynamical systems, model reduction and decompositions," *Nonlinear Dynamics*, vol. 41, no. 1–3, pp. 309–325, 2005.

[27] S. L. Brunton, B. W. Brunton, J. L. Proctor, and J. N. Kutz, "Koopman invariant subspaces and finite-dimensional approximations of nonlinear dynamical systems," *Journal of Nonlinear Science*, vol. 26, pp. 1131–1162, 2016.

[28] B. Lusch, J. N. Kutz, and S. L. Brunton, "Deep learning for universal linear embeddings of nonlinear dynamics," *Nature Communications*, vol. 9, no. 1, p. 4950, 2018.

[29] Y. Liu, C. Li, J. Wang, and M. Long, "Koopa: Learning non-stationary time series dynamics with Koopman predictors," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023, pp. 13 480–13 492.

[30] C. A. Ferro and J. Segers, "Inference for clusters of extreme values," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 65, no. 2, pp. 545–556, 2003.

[31] S. Kim, K. Choi, J. Choi, B. Lee, and S. Yoon, "Towards a rigorous evaluation of time-series anomaly detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 7, 2022, pp. 7194–7201.

[32] R. Wu and E. J. Keogh, "Current time series anomaly detection benchmarks are flawed and are creating the illusion of progress," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 3, pp. 2421–2429, 2023.

[33] M. Bächlin, M. Plotnik, D. Roggen, I. Maidan, J. M. Hausdorff, N. Giladi, and G. Tröster, "Wearable assistant for parkinson's disease patients with the freezing of gait symptom," *IEEE Transactions on Information Technology in Biomedicine*, vol. 14, no. 2, pp. 436–446, 2010.

[34] A. Asuncion and D. Newman, "UCI Machine Learning Repository: CalIt2 Building People Counts Data Set," https://archive.ics.uci.edu/dataset/179/calit2+building+people+counts, 2007.

[35] Y. Zhou, C. Brif, and I. Lourentzou, "mTSBench: A Comprehensive Benchmark for Multi-Domain Time Series Anomaly Detection," *Transactions on Machine Learning Research (TMLR)*, 2026, arXiv:2506.21550.

[36] R. A. Angryk, P. C. Martens, B. Aydin, D. Kempton, S. S. Mahajan, S. Basodi, A. Ahmadzadeh, S. Boubrahimi, S. M. Hamdi, M. A. Schuh *et al.*, "Multivariate time series dataset for space weather data analytics," *Scientific Data*, vol. 7, no. 1, p. 227, 2020.

[37] L. M. Candanedo and V. Feldheim, "Accurate occupancy detection of an office room from light, temperature, humidity and co2 measurements using statistical learning models," *Energy and Buildings*, vol. 112, pp. 28–39, 2016.

[38] S. D. Grimshaw, "Computing maximum likelihood estimates for the generalized pareto distribution," *Technometrics*, vol. 35, no. 2, pp. 185–191, 1993.